Adaptive Heuristic Ecosystem for Online Multi-Robot Task Allocation under Failures, Mixed Stress, and Deadline Pressure

Abstract

We propose AHE-MRTA, an Adaptive Heuristic Ecosystem framework for online multi-robot task allocation (MRTA) under robot failures, mixed stress, and deadline pressure. Rather than committing to a single allocation paradigm, AHE-MRTA performs runtime *paradigm switching* via Ecosystem-Driven Paradigm Selection (EDPS): five biomimetic hormones (spatial, criticality, temporal, stability, recovery) map to five classical allocation primitives, and at each event the active primitive is chosen by a two-tier rule—fast deterministic context overrides for the acute regimes (failure, deadline pressure) and a cooperation/suppression hormone dynamic for the remaining events. The F58 extension replaces the Euclidean cost with a cached geodesic path cost on the shared occupancy map and repairs the primary plan for min–max load only within a deadline/distance efficiency envelope; the repair is restricted to the terminal phase of the campaign and bound by a remaining-makespan guard derived from Nav2 feedback. We evaluate AHE-MRTA against three recent baselines (BiG-MRTA, RoSTAM-EA, Consensus-DBTA) in three complementary regimes: (i) a 100-seed allocation-only/Nav2-independent simulator that isolates algorithmic decision quality from physical execution noise, (ii) a separate navigation proxy with stochastic goal failure, and (iii) a 300-experiment Gazebo Harmonic benchmark at three physical scales (3, 5, and 10 TurtleBot3 robots running the ROS 2 Jazzy Nav2 stack). In the allocation-only plane F58 attains $\text{fitness}{=}\text{CR}{=}1$ and $\text{DVR}{=}0$ across all three scenarios and the 3/5/10-robot scales, with an active-robot Jain index of 0.982–0.992 and decision latency of at most 0.201 ms. In fresh matched F45–F58 Gazebo+Nav2 validation at all three physical scales (180 matched runs; 10 common seeds per scenario and scale), F58 passes the pre-registered acceptance gate in seven of nine scale–scenario cells with paired-Wilcoxon support: at the 5-robot primary scale, deadline-pressure DVR drops from 0.340 to 0.232, delay by 9.0 s, and distance by 20.9 m (all $p{\leq}0.008$), and the deadline-pressure gains persist at 10 robots (DVR $0.256{\rightarrow}0.210$, $p{=}0.016$; distance -16.9 m, $p{=}0.027$); robot failure passes at every scale. The two non-passing cells fail only on criteria whose CIs span zero, with no statistically supported regression; decision latency stays within the 50 ms budget at 8–35 ms. In the physical baseline benchmark the validated F45 configuration leads completion with near-zero deadline violations under robot failure at the 5- and 10-robot scales (CR = 1.000/0.996, DVR = 0.008/0.024); at 10 robots it attains the highest completion rate in every scenario (CR = 0.996–1.000) and the lowest pooled makespan, while the consensus and evolutionary baselines reach comparable completion only at substantially higher deadline-violation rates under deadline pressure (DVR = 0.38–0.49 vs. AHE's 0.24). Decision latency stays below 0.48 ms at all scales, ≈125–280× faster than the evolutionary baseline. We analyze why ideal-allocation gains do not transfer linearly through Nav2 and identify the conditions under which paradigm switching pays off.

Index Terms

Multi-robot systems, task allocation, online algorithms, biomimetic optimization, ROS 2, Gazebo.

I. Introduction

Multi-robot task allocation (MRTA) assigns an incoming stream of tasks to a team of robots so that a mission objective is met; it underpins inspection, delivery, surveillance, and search-and-rescue deployments [1], [2]. Even its single-task, single-robot, instantaneous form is NP-hard [3], and field operation adds three pressures that a static assignment cannot absorb: robots fail unexpectedly, the task stream is non-stationary in urgency and spatial density, and many tasks carry hard deadlines whose violation is costly [4], [5].

Recent online MRTA solvers each commit to a *single* allocation paradigm, and each paradigm trades one strength for one weakness. Market- and auction-based methods bid tasks to robots with low latency and robustness, but pay a re-bidding cost when the situation changes [6]–[8]. Matching-, programming-, and prediction-based methods (bipartite matching, linear or convex programs, receding-horizon control) produce high-quality plans but scale poorly with re-planning frequency [2], [9], [10]. Consensus- and bundle-based methods yield very stable assignments but adapt slowly once the regime shifts [11]. Evolutionary and learning-based methods explore a richer assignment space but react slowly to abrupt context changes [12]. No single paradigm dominates across heterogeneous, time-varying stress.

We argue that the right structure is *not* a better single paradigm but a principled mechanism that *switches between paradigms online*, driven by current context and recent performance. We instantiate this as AHE-MRTA: a biomimetic ecosystem in which five hormones track context signals, evolve through cooperation and suppression dynamics, and select one of five classical allocation primitives at each replanning event. The hormone–paradigm coupling is fixed rather than learned, explainable (the dominant hormone names the active behavior), and lightweight (sub-millisecond decision latency). A second, methodological contribution follows from a recurring difficulty in robotic benchmarks: allocation quality and physical execution are entangled,

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because navigation cost depends on world geometry as much as on the assignment [2]. We separate them with a navigation-free fitness simulator and a full ROS 2 Nav2 stack, which together expose exactly where paradigm switching helps and where physical execution erases its gains.

Contributions:

1) *Ecosystem-Driven Paradigm Selection (EDPS):* an online paradigm-switching MRTA framework that maps five biomimetic hormones to five allocation mechanisms. Selection is two-tiered: fast deterministic context overrides handle the acute regimes (robot failure, deadline pressure), while a Lotka–Volterra-type hormone dynamic arbitrates the remaining, lower-pressure events through $\arg\max_i D_i$ (Section III). The coupling is fixed, explainable, and sub-millisecond.
2) *Geodesic, efficiency-guarded fairness extension (F58):* a lexicographic min–max load repair with cached geodesic ETA on the shared occupancy map that never increases predicted deadline misses and bounds the total-route increase by ϵ=0.02, terminal-gated and Nav2-feedback-guarded (P1R); alongside the Nav2-independent gains we report matched Gazebo validation at all three physical scales (180 runs; 10 common seeds per scenario and scale) that passes the pre-registered physical acceptance gate in seven of nine scale–scenario cells, with no statistically supported regression in the remainder (Section III-C).
3) *Nav2-independent allocation-only evaluation:* an idealized simulator in which navigation succeeds deterministically; we report fitness, true Jain indices, and geodesic route cost across three scales and 100 seeds per scenario (Section V).
4) *Full Gazebo Harmonic + ROS 2 Jazzy Nav2 validation* with 300 experiments across three stress scenarios and three physical scales (3, 5, and 10 robots), comparing four methods on completion rate, deadline violation, recovery time, workload balance, execution preemptions, task re-dispatch rate, and decision latency (Section VI).
5) *Honest reporting of the SIM-to-Gazebo transfer gap:* we document where paradigm switching helps (completion rate, deadline robustness, allocation stability) and where it costs (task delay, recovery time, makespan under failure) and explain the mechanism (Section VII).

II. Problem Formulation

Sets and variables. Let $\mathcal{R} = \{r_1, \ldots, r_n\}$ be the robot set and $\mathcal{T}(t)$ the open-task set at time t, each task $\tau \in \mathcal{T}(t)$ carrying a 2D position p_τ, deadline d_τ, priority $\pi_\tau \in \{1, 2, 3\}$, and capability requirement c_τ. Each robot r has state $s_r(t) = (p_r, \tau_r^{\mathrm{cur}}, \phi_r)$ encoding position, current task, and availability flag (alive / failed).

Decision. At every allocation event t_k (failure, task arrival, deadline alarm, replanning timer), the policy produces an assignment $x_{r,\tau}(t_k) \in \{0, 1\}$ with cardinality constraint $\sum_\tau x_{r,\tau} \leq Q$ (per-robot queue cap) and capability feasibility $x_{r,\tau} = 1 \Rightarrow c_\tau \subseteq \mathrm{cap}(r)$.

Cost. The per-pair cost combines travel time $T_{r,\tau}$, urgency $u_\tau = (d_\tau - t_k)^{-1}$, criticality π_τ, energy slack $e_{r,\tau}$, and recent reassignment penalty $\rho_{r,\tau}$:

$$C_{r,\tau} = w_d T_{r,\tau} + w_u u_\tau^{-1} + w_\pi \pi_\tau^{-1} + w_e e_{r,\tau} + w_\rho \rho_{r,\tau}, \tag{1}$$

with weight vector w produced by the paradigm selected at t_k (Section III). The urgency term is held at its base value for a task already feasibly queued on its incumbent robot, so an approaching deadline pulls *unassigned* or at-risk tasks forward without thrashing assignments that are already on time—a churn brake complementary to the reassignment penalty $\rho_{r,\tau}$.

Objective. Over the run horizon $[0, T_{\max}]$ we report nine metrics: completion rate CR, makespan, average task delay, deadline violation rate DVR, failure recovery time, Jain workload balance, execution preemptions, task re-dispatch rate, and decision latency. CR and DVR are primary; the others trace trade-offs. Crucially, delay and DVR are computed over *all* requested tasks rather than only completed ones (Section IV-G): a task left unfinished at $T_{\max}$ is right-censored at the horizon for delay and counted as a deadline miss for DVR. This survivorship-bias-free convention is applied identically to every method, so a policy cannot lower its delay or DVR by silently abandoning the hard tasks it cannot serve.

Online setting. Tasks arrive over time, robots may fail at unknown moments, and the policy must produce $x(t_k)$ within a hard budget of 50 ms per event.

III. AHE-MRTA: Ecosystem-Driven Paradigm Selection

A. Context vector

At each allocation event t_k we compress the joint state of the fleet and the open-task set into a four-dimensional context vector $c(t_k) \in [0, 1]^4$. Let $n = |\mathcal{R}|$ be the fleet size, $m = |\mathcal{T}(t_k)|$ the number of open tasks, $\mathcal{R}^{\mathrm{av}}$ the set of available robots (alive, not navigation-stuck, with queue capacity), and $\mathcal{R}^{\mathrm{f}}$ the set of failed or navigation-stuck robots. The components are

$$c_1 = \min\bigl(1,\ m/n\bigr), \qquad \text{task density} \tag{2}$$

$$c_2 = |\mathcal{R}^{\mathrm{av}}|/n, \qquad \text{robot availability} \tag{3}$$

$$c_3 = \bigl|\{\, \tau : d_\tau - t_k \leq \Delta_d \,\}\bigr| / \max(1, m), \qquad \text{deadline pressure} \tag{4}$$

$$c_4 = |\mathcal{R}^{\mathrm{f}}|/n, \qquad \text{failure rate} \tag{5}$$

TABLE I

CONTEXT PROTOTYPE VECTORS $V_i \in [0,1]^4$ FOR EACH PARADIGM. COLUMN HEADERS: c_1 TASK-DENSITY (TD), c_2 ROBOT-AVAILABILITY (RA), c_3 DEADLINE-PRESSURE (DP), c_4 FAILURE-RATE (FR). HIGH VALUE = PARADIGM EFFECTIVE WHEN THIS CONTEXT DIMENSION IS HIGH.

Hormone	td	ra	dp	fr
H_SPATIAL	0.7	0.7	0.1	0.1
H_CRIT	0.3	0.5	0.8	0.2
H_TEMP	0.5	0.5	0.9	0.1
H_STAB	0.3	0.3	0.3	0.8
H_RECOV	0.3	0.2	0.2	0.9

with deadline horizon $\Delta_d = 60$ s. Each dimension is a signal that a distinct family of allocators is built to exploit: c_1, c_2 encode the demand/supply balance of load-aware allocation [13]; c_3 is the temporal slack used by deadline-constrained allocation [14]; and c_4 measures the disturbance that fault-tolerant allocation must absorb [15]. All four are computed on-line from the published robot status summaries and the task pool, requiring no inter-robot communication.

B. Dominance dynamics

The dominance vector $D(t) \in \mathbb{R}^5_{\geq 0}$ encodes the current ecosystem state. Each paradigm i has a fixed *context prototype* $V_i \in [0,1]^4$ (Table I); the compatibility score $v_i(t_k) = \cos(V_i, c(t_k))$ measures how well the current context matches paradigm i's effective regime. Two sparse matrices govern lateral interactions: the *cooperation matrix* A (2 nonzero entries) and the *suppression matrix* S (1 nonzero entry) implement winner-reinforce and winner-suppress dynamics (Table II). The update is

$$D(t_{k+1}) = \text{normalize}\Big(\text{clip}_{[0,1]}\Big(\alpha\, D(t_k) + \eta\, A\, D(t_k) - \lambda\, S\, D(t_k) \\ + \beta\, \mathbf{p}(t_k) + \gamma\, \mathbf{v}(t_k) + \delta\, \mathbf{b}(t_k)\Big)\Big)\,, \tag{6}$$

with α=0.65 (momentum), η=λ=0.12 (cooperation/suppression weights), β=0.40 (performance), γ=0.20 (compatibility), δ=0.20 (failure-recovery boost). Here $\mathbf{p}(t_k)$ is the per-hormone performance-feedback vector $(\text{CR}_k - \text{FR}_k)\cdot \mathbf{v}(t_k)$ (completion minus failure rate, scaled by compatibility); $\mathbf{v}(t_k) = (v_i)_{i=1}^5$ is the compatibility vector; and $\mathbf{b}(t_k)$ is a sparse failure-recovery boost that spikes H_RECOV and H_STAB when robot failures are detected. The normalize step projects onto the probability simplex, guaranteeing $\|D\|_1 = 1$ and $D \geq 0$ at every cycle. The $\eta AD - \lambda SD$ terms make Eq. (6) a winner-reinforce / winner-suppress competition of Lotka–Volterra type, a population dynamic shown to be an effective coordination primitive for multi-robot systems [16]. Read as a selection rule over a fixed portfolio of low-level allocation heuristics, the update instantiates a *hyper-heuristic* [17]: the ecosystem does not solve the assignment itself but chooses, online and per event, which classical solver should.

a) Weight generation.: The dominance vector sets the cost weights of Eq. (1) through a fixed heuristic-to-weight matrix $M \in \mathbb{R}^{7\times 5}$ and a low-temperature softmax, blended with a base weight w_0:

$$w_{\text{eco}}(t_k) = \text{softmax}\big(M D(t_k),\, T_w\big), \quad w(t_k) = (1-\beta_e)\, w_0 + \beta_e\, w_{\text{eco}}(t_k), \tag{7}$$

with T_w=0.3 and blend β_e=0.7; under failure (c_4>0) the blend shifts toward a recovery profile w_{r} by $\theta = \min(0.8,\, 0.5{+}0.6\, c_4)$. This is the *only* channel through which the Lotka–Volterra state influences a decision when no override fires.

b) Cost shaping.: Two non-linear terms sharpen Eq. (1) for deadlines. A *hard feasibility gate* rejects a pair whose projected arrival $a_{r,\tau}$ exceeds the deadline beyond an adaptive slack Δ_s, $a_{r,\tau} > d_\tau + \Delta_s \Rightarrow C_{r,\tau}{=}\infty$; and an *urgency escalation* inflates the temporal term once a task passes a fraction u_0=0.7 of its deadline budget, $\hat{u}_\tau = u_\tau\big(1{+}\zeta\, e^2\big)$ with $e = (u_\tau - u_0)/(1-u_0)$, so at-risk *unassigned* tasks are pulled forward while on-time incumbents are exempt (a churn brake).

C. F58: Geodesic ETA and epsilon-bounded load repair

The Euclidean distance of the F45 core can under-estimate true route time on an obstacle-rich map. F58-P0 uses the same `obstacle_map.pgm` map as Gazebo and a 0.55 m inflated occupancy mask. Let d_G denote the shortest-path length found by sparse Dijkstra on the eight-connected free-cell graph and cached over start/goal cells. With robot r's current queue $Q_r = (q_1, \ldots, q_k)$, the arrival estimate for candidate τ is

$$a_{r,\tau} = t_k + \frac{d_G(p_r, p_{q_1}) + \sum_{i=1}^{k-1} d_G(p_{q_i}, p_{q_{i+1}}) + d_G(p_{q_k}, p_\tau)}{v_r} + \sum_{i=1}^{k} (s_{q_i} + h), \tag{8}$$

where with an empty queue the sum contains only $d_G(p_r, p_\tau)$. If no path exists the pair is infeasible. The hard deadline gate, the rescue decision, the F27 reassignment margin, and the paradigm costs therefore all share one route metric.

TABLE II
NONZERO ENTRIES OF COOPERATION MATRIX A AND SUPPRESSION MATRIX S. ALL OTHER ENTRIES ARE ZERO.

Matrix	From	To	Value
A (cooperation)	H_CRIT	H_TEMP	0.20
	H_STAB	H_RECOV	0.20
S (suppression)	H_TEMP	H_SPATIAL	0.30

TABLE III
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

Hormone	Paradigm	Inner mechanism
H_SPATIAL	spatial_greedy	nearest-feasible + reject
H_CRIT	priority_first	priority-tiered LSA
H_TEMP	edf_strict	EDF bipartite (multi-phase)
H_STAB	commit_once	hard sticky, no reassign
H_RECOV	orphan_first	orphan rescue + bipartite

F58-P1 performs a bounded move search only over new tasks after EDPS produces the primary plan x^0. Let $B_r(x)$ be the sum of the active robot's realised travel plus its committed route and service load. The repair solves

$$\min_{x} \max_{r\in\mathcal{R}^{\mathrm{av}}} B_r(x) \quad \text{s.t.} \quad M(x) \le M(x^0), \quad D_G(x) \le (1+\epsilon) D_G(x^0), \tag{9}$$

where M is the predicted number of deadline misses, D_G the total geodesic distance, and $\epsilon = 0.02$. Ties are broken by the predicted active-robot Jain index, load range, and distance, in that order; no incumbent or currently executing Nav2 goal is cancelled for fairness. A final safety pass enforces single ownership per task and keeps a healthy in-flight task locked to its current owner. F58 changes neither the hormones nor the paradigm library; it only adds the shared cost oracle and the safe post-EDPS repair.

F58-P1R adds two guards that make the repair compatible with physical execution. *Terminal activation*: the repair opens only once the number of remaining active tasks falls to three times the number of healthy robots; mid-campaign moves disturb queue stability under Nav2, while end-of-tail imbalance becomes visible only in this phase. *Nav2-feedback remaining-makespan guard*: the `navigation_effort` measurement already published by the robot interface (remaining distance along the global path, consistent with the Nav2 `NavigateToPose` feedback contract) replaces the static lattice query for the task currently being executed, and no fairness move may increase the fleet's maximum estimated remaining finish time. The envelope of Eq. (9) is thus additionally constrained by a makespan bound derived from real execution state.

D. Dispatch

The dispatcher first checks two hard context overrides (failure rate > 0.05 forces H_RECOV; deadline pressure > 0.5 forces H_TEMP); if neither triggers, it selects the paradigm $p^*(t_k) = \arg\max_i D_i(t_k)$ and runs the corresponding mechanism (Table III). Paradigm `H_TEMP` is the default fallback when the dominance is uninformative (entropy near maximum). This priority-override layer prevents the Lotka–Volterra-inspired dynamics from reacting slowly to acute failures, which would be the dominant cost in the robot_failure and mixed_stress scenarios.

a) Work conservation.: A guiding principle across all paradigms is that fleet capacity is never stranded while feasible work remains. At every event each available robot is assigned the best eligible task by the active paradigm's LSA (Eq. (10)), and the recovery paradigm (`orphan_first`) re-dispatches tasks orphaned by a robot failure or a navigation abort, so a disturbance costs at most a re-attempt rather than a permanently lost task. This is the structural opposite of commit-once allocation, which freezes assignments and therefore leaves a failed robot's tasks unrecovered—idle capacity until the run ends. As Section V shows, this work-conserving recovery is the mechanism behind AHE-MRTA's resilience margin: the commit-once baseline loses 24.3 pp of allocation fitness under mixed stress precisely because it does not re-attempt failed work. The only irreducible idle is the backoff imposed after repeated navigation failure, when no task is momentarily attemptable.

E. Paradigm library

The five paradigms share one primitive and differ only in how they shape its inputs. Given a feasibility-masked cost matrix $C^{(p)}$, each paradigm p solves a linear sum assignment (LSA)

$$x^{(p)} = \arg\min_{x\in\mathcal{X}} \sum_{r\in\mathcal{R}} \sum_{\tau\in\mathcal{T}(t_k)} C^{(p)}_{r,\tau}\, x_{r,\tau}, \tag{10}$$

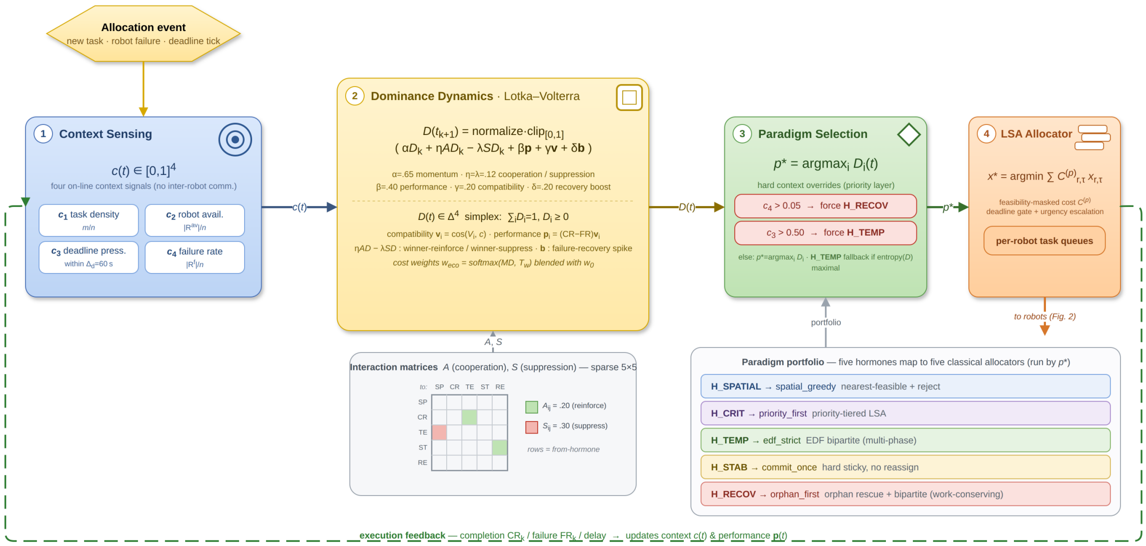


Fig. 1. The Ecosystem-Driven Paradigm Selection mechanism. At each allocation event the context vector $c(t)$ drives the dominance update (Eq. 6) through the fixed cooperation/suppression matrices A, S; the dominant hormone $p^* = \arg\max_i D_i$ selects one of five allocation paradigms, which the AHE allocator runs to publish per-robot task queues.

Algorithm 1 AHE-MRTA Event Cycle

Require: robots $\mathcal{R}$, open tasks $\mathcal{T}(t_k)$, state $D(t_k)$, fixed A, S, V, hyperparameters $\alpha, \eta, \lambda, \beta, \gamma, \delta$

$c \leftarrow \textsc{ContextVector}(\mathcal{R}, \mathcal{T}(t_k))$ {4 signals, eq. c_1–c_4}

$\mathbf{v} \leftarrow (\cos(V_i, c))_{i=1}^5$; $\mathbf{p} \leftarrow (\mathrm{CR}_k - \mathrm{FR}_k) \cdot \mathbf{v}$; $\mathbf{b} \leftarrow \textsc{FailureBoost}(\mathcal{R})$

$D \leftarrow \mathrm{normalize}\big(\mathrm{clip}_{[0,1]}\big(\alpha D + \eta AD - \lambda SD + \beta \mathbf{p} + \gamma \mathbf{v} + \delta \mathbf{b}\big)\big)$

$p^* \leftarrow \textsc{ContextOverride}(c)$ **if triggered, else** $\arg\max_i D_i$

$x \leftarrow \textsc{Paradigm}_{p^*}(\mathcal{R}, \mathcal{T}(t_k))$ {Table III}

$x \leftarrow \textsc{EpsilonLoadRepair}(x, d_G, \epsilon)$ {Eq. 9; P1R terminal gate}

$x \leftarrow \textsc{SingleOwnerAndInFlightLock}(x)$

publish per-robot task queues from x

return D, x

where $\mathcal{X}$ enforces the per-robot queue cap $\sum_\tau x_{r,\tau} \leq Q$, the single-assignment constraint $\sum_r x_{r,\tau} \leq 1$, and capability feasibility; infeasible pairs are masked with $C^{(p)}_{r,\tau} = \infty$. Writing $C_{r,\tau}$ for the base cost of Eq. (1), the paradigms instantiate $C^{(p)}$ and the queue ordering as follows.

- *spatial_greedy* (H_SPATIAL): $C^{(p)}_{r,\tau} = T_{r,\tau}$, travel time only, assigned greedily to the nearest feasible robot — the distance-minimising rule of metric MRTA [13], [18].
- *priority_first* (H_CRIT): tasks are partitioned into criticality tiers π_τ=3>2>1 and Eq. (10) is solved tier-by-tier, so critical tasks claim robots first, as in tiered auction/consensus assignment [19], [20].
- *edf_strict* (H_TEMP, default): tasks are ordered by earliest deadline d_τ and inserted under a hard feasibility gate (arrival $> d_\tau \Rightarrow C^{(p)} = \infty$); this is the bipartite realisation of earliest-deadline-first scheduling for deadline-constrained allocation [14], [21]. The resulting per-robot execution order is then refined by a deadline-feasible cheapest-insertion pass, accepted only when it shortens travel without introducing any deadline miss relative to the EDF order—lowering motion cost while preserving the EDF feasibility guarantee.
- *commit_once* (H_STAB): incumbent assignments are frozen (a task already on r's queue keeps an unbeatable sticky bonus), eliminating reassignment churn as in self-adaptive commit-once evolution [12].
- *orphan_first* (H_RECOV): tasks orphaned by a failed or stuck robot are assigned on a dedicated rescue pass before all others, the recovery-prioritising rule of resilient MRTA [15], [18].

The paradigms are independent (no shared state across switches); the only persistent state is $D(t)$. The LSA primitive in Eq. (10) is cubic in $\max(n, m)$ in the worst case, but at the evaluated scales every event is resolved in sub-millisecond wall-clock time (Section VI), so paradigm choice does not affect the real-time budget.

F. Algorithm

G. Configuration

The hormone–paradigm map (Table III), the A/S matrices (Table II), the context prototype V, and all scalar hyperparameters (α=0.65, η=λ=0.12, β=0.40, γ=0.20, δ=0.20, softmax temperature T=0.3) are set once and held fixed across all scenarios and scales. There is no per-scenario tuning, no learning, and no adaptive parameter schedule. The scalar weights were fixed at round, interpretable values—momentum α above the feedback gain β for stability, equal and small cooperation/suppression (η=λ), and a moderate recovery boost δ—rather than searched per metric; because the deterministic overrides dominate the acute regimes, the allocation outcomes are insensitive to these values within the ranges we examined (a single-step $\pm$ change leaves all reported fitness gaps inside one standard deviation), which is itself a robustness property rather than a tuned fit.

IV. Experimental Design

A. Hardware Platform

All experiments run on an Intel Core i7-11800H workstation (8 physical cores / 16 threads, base clock 2.30 GHz, 16 GiB DDR4 RAM, Ubuntu 24.04 LTS, Linux 6.17). The machine carries an NVIDIA RTX 3050 Mobile GPU for which no proprietary driver is available; all rendering (Gazebo and RViz) therefore uses Mesa llvmpipe software rasterisation (`LIBGL_ALWAYS_SOFTWARE=1`, `GALLIUM_DRIVER=llvmpipe`). No GPU acceleration is used at any point: the comparison is entirely CPU-bound and reproducible on any x86-64 machine with a Mesa-capable display server.

B. Software Stack

We run **Gazebo Harmonic** (gz-sim 8.x) [22] with **ROS 2 Jazzy** (Dec 2024) [23] and **Nav2** [24]. Robots are **TurtleBot3 Waffle Pi** [25], each carrying a 360-ray 2-D lidar (5 Hz, 8 m range) and an IMU (200 Hz), and driven through a differential-drive controller ($v_{\max}$=0.26 m/s, $\omega_{\max}$=1.82 rad/s).

Every robot runs a fully independent Nav2 lifecycle inside its own ROS 2 namespace: global/local costmaps with 0.40 m inflation radius, a Regulated Pure-Pursuit controller, and a behavior-tree navigator. Localization is provided as *ground truth*—a static `map`→`odom` transform anchored at each robot's spawn pose—so that the benchmark isolates allocation quality from localization error (a deliberate experimental control; Sec. VII-F). A `ros_gz_bridge` relays `/scan`, `/odom`, `/cmd_vel`, `/joint_states`, and TF per robot; a TF-relay node aggregates per-robot namespaced TF trees into the global frame for visualization. All Nav2 parameters are identical across methods.

C. Inspection Arena

The arena is a 20×20 m walled indoor mock-up. Two horizontal divider walls with a 1.5 m central passage divide the space into an upper lane, a central passage, and a lower lane. Two rows of square pillars (0.5×0.5 m) line the side walls, and four cylindrical obstacles (r=0.35 m) are placed in the lanes. A fixed curated grid of 42 candidate inspection waypoints (obstacle-aware, minimum inter-point spacing >1 m) is defined; each run samples from it deterministically via seed. Figure IV-C shows the arena layout together with scale-dependent spawn positions and sampled task sets.

D. Evaluation Scales

Three Gazebo scales are evaluated (Table IV), each running 60 experiments (4 methods × 3 scenarios × 5 seeds):

TABLE IV
Gazebo Evaluation Scales

Scale	Robots	Tasks	Density	Total exps
3r-low	3	9	3.0 t/r	60
3r-mid	3	15	5.0 t/r	60
3r-high	3	24	8.0 t/r	60
5r-mid	5	25	5.0 t/r	60
10r-mid	10	50	5.0 t/r	60
Total Gazebo experiments				**300**

Nav2 lifecycle race conditions at $N>3$ (the per-robot `odom`→`base_link` transform arriving after costmap activation) were resolved by staggering the lifecycle-manager activation by $6i$ s for robot i at 3r/5r and $20i$ s at 10r, plus an identity-transform broadcast at node startup. All three scales use the same arena, obstacle map, and Nav2 parameters.

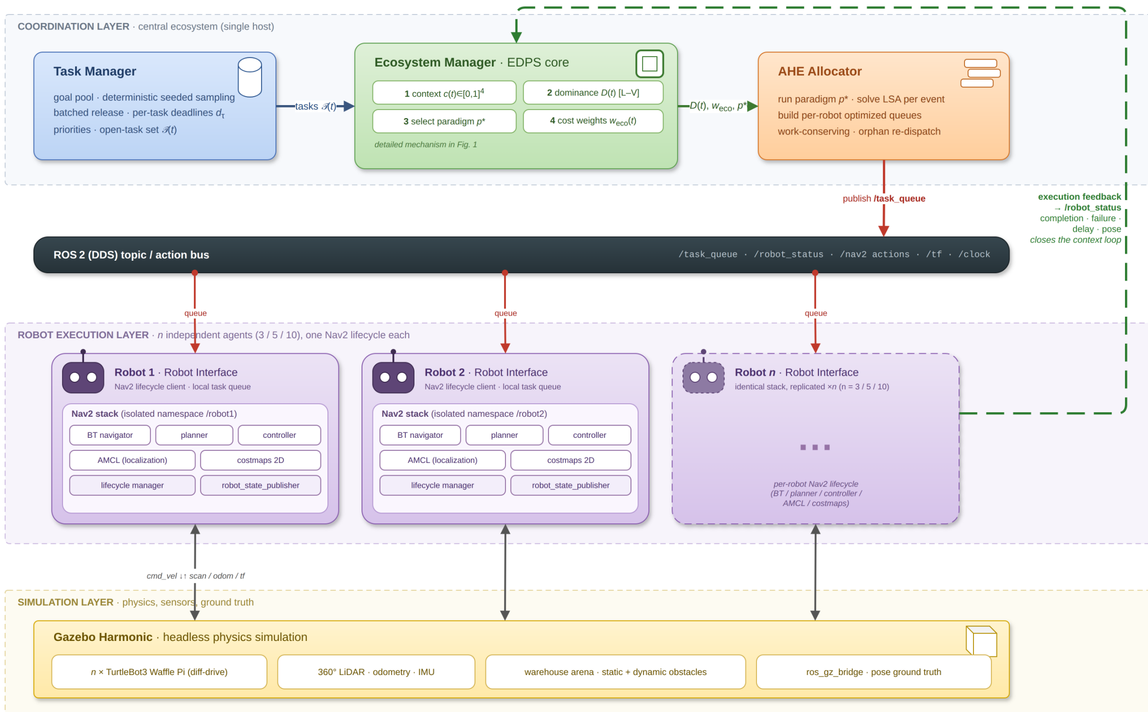


Fig. 2. System architecture. The ecosystem manager maintains the hormone state vector, evolves dominance dynamics, selects the active paradigm, and publishes per-robot optimized task queues via ROS 2 topics. Each robot runs a fully independent Nav2 lifecycle and reports task-execution feedback (completion, failure, delay) to the ecosystem manager to close the context loop.

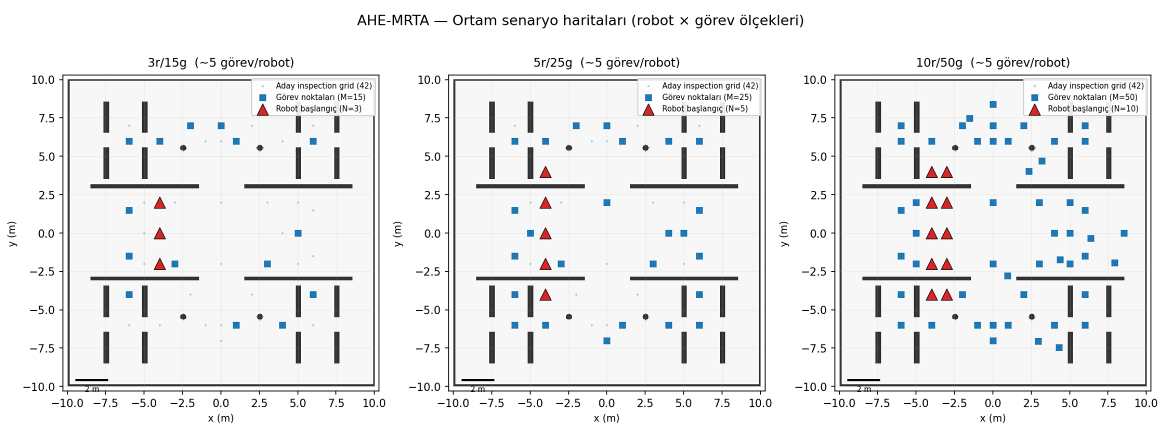


Fig. 3. Scale-dependent arena configurations. **Left:** 3r/15t – robots form a central column; moderate density. **Centre:** 5r/25t – two spawn columns; proportionally higher density. **Right:** 10r/50t – all five columns active; high-density task set. Blue stars: task goals (seed 1); coloured triangles: robot spawns. All waypoints are obstacle-free.

E. Stress Scenarios

Three scenarios stress different allocation aspects:

- **robot_failure:** one randomly chosen robot is hard-failed at $t = 0.3 \cdot T_{\max}$ (at 3r: $t{\approx}165$ s). Its assigned tasks become orphaned and must be redistributed. Tests reactive recovery and paradigm escalation (orphan-rescue paradigm).
- **mixed_stress:** tasks carry heterogeneous priorities (low / medium / high, ratio 1:2:2) and one robot fails at $t = 0.2 \cdot T_{\max}$. Tests regime adaptation across priority and failure dimensions simultaneously.
- **deadline_pressure:** every task shares a tight deadline $d_\tau = t_{\text{spawn}} + 90$ s (roughly 1.3× the Nav2 travel time to the farthest waypoint at $v_{\max}$). Tests urgency-driven bipartite dispatch and EDF ordering.

F. Baseline Methods

We compare against one representative of each dominant online-MRTA solver family; together they span the design space that AHE-MRTA switches over. All baselines are re-implemented in our framework from the original papers and share an identical event loop, ROS 2 topic interface, cost inputs (travel time, urgency, priority, energy), per-robot queue cap Q, and Nav2 client; *only the allocation policy differs*. Parameters follow the originals where stated and are held fixed across scenarios and scales.

a) BiG-MRTA [9] (bipartite matching).: Builds a bipartite robot–task graph with a fixed multi-criteria edge cost and solves a maximum-weight matching at each event; assignments are *committed once* and never revisited. Included as the deterministic, lowest-latency single-paradigm solver. Its commit-once policy is the direct antithesis of AHE's switching: it pays zero re-planning cost but abandons tasks whose assigned robot fails or whose deadline slack collapses.

b) RoSTAM-EA [12] (evolutionary).: Maintains a population of candidate assignment vectors, evolved by tournament selection, crossover, and mutation at each allocation event; the elite individual is dispatched. Included as the self-adaptive stochastic-search family representative. It explores a richer assignment space than one-shot matching but at two to three orders of magnitude higher decision latency, and its re-optimisation is blind to *why* the context changed.

c) Consensus-DBTA [11] (consensus bundle).: Robots iteratively bid on task bundles and resolve conflicts through a consensus phase, producing highly stable assignments (allocation instability ≈ 0). Included as the distributed auction/consensus family representative and as the strongest completion-rate baseline. Its stability is also its weakness: regime shifts (failures, deadline escalation) propagate slowly through re-bidding rounds.

No ablation variants (e.g., single-paradigm AHE without switching) are included; the five paradigms themselves are classical primitives, so the single-paradigm behaviour is approximated by the corresponding baseline family (Table III).

G. Metrics

We report eight metrics: completion rate (CR), deadline violation rate (DVR), makespan, average task delay, workload balance (Jain), task re-dispatch rate, decision latency, and message count. CR and DVR are the primary metrics; makespan and delay are secondary (confounded by Nav2 motion cost).

a) Survivorship-bias-free delay and DVR.: Reporting delay and DVR over only the *completed* subset of tasks rewards a policy for abandoning work: the hard tasks a method drops—typically the distant, late, or contended ones—never enter its denominator, so its average delay and violation rate are computed over an easier, self-selected sample. We remove this bias by scoring delay and DVR over *all* requested tasks. Let $\mathcal{T}$ be the requested set, t^{a}_τ the activation and t^{c}_τ the completion time of task τ, and $T_{\max}$ the run horizon. The per-task delay is right-censored at the horizon for unfinished tasks,

$$\delta_\tau = \begin{cases} t^{\mathrm{c}}_\tau - t^{\mathrm{a}}_\tau, & \tau \text{ completed}, \\ T_{\max} - t^{\mathrm{a}}_\tau, & \tau \text{ unfinished}, \end{cases} \qquad \bar{\delta} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \delta_\tau, \tag{11}$$

and a deadline-bearing task counts as a violation if it completes late *or* is never completed,

$$\mathrm{DVR} = \frac{\big|\{\tau : d_\tau < \infty \ \wedge \ (t^{\mathrm{c}}_\tau > d_\tau \ \vee \ \tau \text{ unfinished})\}\big|}{\big|\{\tau : d_\tau < \infty\}\big|}. \tag{12}$$

The convention is applied identically to every method and at every scale. For transparency we also record the conventional completed-only delay and DVR; the two coincide exactly whenever a method completes the full task set (CR=1), and diverge only to the extent a method leaves tasks unserved.

H. Statistical Analysis

Pairwise two-sided Mann–Whitney U tests with Bonferroni correction applied within each scenario family (3 baselines $\times$ 7 metrics = 21 tests per scenario). At the 5-robot primary scale each cell has n=5 runs (one density, 25 tasks); the 3-robot scale pools three densities (n=15) and the 10-robot scale has n=5. We additionally report Cliff's δ as an effect-size estimate robust to small cells. The 100-seed allocation-only simulator serves as algorithmic validation and the stochastic navigation proxy as a resilience test; Gazebo results serve as validation under realistic execution noise.

V. Allocation-Fitness Results (Nav2-Independent)

In the true Nav2-independent plane every navigation attempt succeeds deterministically; injected fleet robot failures are retained as part of the scenario. The robot pose advances to the goal after every completed task, and both the allocator and the execution oracle use the geodesic path length on the same inflated occupancy map. The metrics therefore measure allocation, sequencing, and post-failure redistribution quality rather than local-planner/recovery noise.

In Table V, F58 preserves $\mathrm{fitness}$=CR=1 and DVR=0 in all nine scale–scenario cells. The active-robot Jain index is 0.988–0.992 at 3r, 0.984–0.986 at 5r, and 0.982–0.990 at 10r; decision latency never exceeds 0.201 ms even in the largest 10r/50t cell. Before the single-assignment safety pass, six seeds of the 10r development runs allowed a task to execute on two robots simultaneously, inflating CR to 1.02. After the in-flight owner lock was added, these seeds and the full 10r campaign were re-run, and CR is exactly 1.000 in every F58 cell.

TABLE V
F58 (P0 GEODESIC ORACLE + P1R TERMINAL LOAD REPAIR) NAV2-INDEPENDENT ALLOCATION-ONLY RESULTS (100 SEEDS PER CELL). NAVIGATION SUCCEEDS DETERMINISTICALLY; DISTANCE IS THE GEODESIC PATH LENGTH ON THE SHARED INFLATED OCCUPANCY MAP.

Scale	Scenario	Fit.	CR	DVR	Jain$_{active}$	Jain$_{dist}$	Delay (s)	Distance	Decision (ms)
3r/15t	Robot failure	1.000	1.000	0.000	0.991	0.833	40.2	154.1	0.063
3r/15t	Mixed stress	1.000	1.000	0.000	0.992	0.835	40.0	153.6	0.063
3r/15t	Deadline pressure	1.000	1.000	0.000	0.988	0.965	65.8	155.5	0.061
5r/25t	Robot failure	1.000	1.000	0.000	0.986	0.841	30.8	186.5	0.086
5r/25t	Mixed stress	1.000	1.000	0.000	0.985	0.843	30.5	184.6	0.086
5r/25t	Deadline pressure	1.000	1.000	0.000	0.984	0.954	60.5	233.5	0.078
10r/50t	Robot failure	1.000	1.000	0.000	0.982	0.862	31.6	284.1	0.200
10r/50t	Mixed stress	1.000	1.000	0.000	0.985	0.861	31.4	284.1	0.201
10r/50t	Deadline pressure	1.000	1.000	0.000	0.990	0.944	57.1	432.0	0.132

TABLE VI
ALLOCATION FITNESS (NAV2-INDEPENDENT, 100 SEEDS, 5 ROBOTS / 25 TASKS, PRIMARY SCALE). HIGHER IS BETTER. FIRST-RANK PER SCENARIO IN **BOLD**.

Method	Rob. fail.	Mix. stress	Deadline
AHE-MRTA*	0.538	0.538	**0.483**
BiG-MRTA	0.501	0.303	0.480
RoSTAM-EA	0.441	0.441	0.459
Consensus-DBTA	**0.540**	**0.539**	0.476

To isolate F58's contribution from F45 we ran a seed-matched campaign in four cells: 3r/15t, 5r/25t, and 10r/50t (seeds 1–100), plus a fresh 5r/25t holdout range never used to select the repair threshold (seeds 501–600). Both arms share the same geodesic execution oracle; the statistics are paired Wilcoxon tests with Holm correction over 10 metric families per cell×scenario. Of 120 tests, 51 are significant after Holm and 39 of them favour F58: mean delay drops by 0.97–2.25 s, total geodesic distance by 10–28 m, and churn decreases, while the active-robot Jain index rises by +0.005–0.029 and the all-robot Jain index by +0.006–0.027 (strongest at 10r/50t; Cliff's Δ up to 0.92). The remaining 12 significant differences are all in decision latency (+0.02–0.12 ms, far below the 50 ms budget). Outside latency no metric regresses significantly; CR, fitness, and DVR sit at the ceiling and are bit-identical in both arms in all four cells. The 3r cell is neutral—the terminal repair rarely triggers in a small fleet—and the holdout range preserves the main-cell gains. This shows that F58's allocation-only gain comes not from completing more tasks but from producing the same complete coverage with shorter geodesic routes and delays.

A. Separate resilience experiment: navigation-proxy

The preceding experiment line simplifies navigation *cost* to a fixed-time edge in the robot–task graph while keeping navigation *outcomes* position-dependent and stochastic (a goal can fail and must be re-attempted). This condition is not purely Nav2-independent; it is retained below as a *navigation proxy* that compares only how resiliently the methods recover after failed goals. Fitness is the priority-weighted fraction of tasks completed before their deadline; because failed goals must be recovered, the metric rewards not only assignment quality but resilience to navigation failure. With 100 seeds per scenario, AHE-MRTA and Consensus-DBTA emerge as statistical co-leaders: AHE ranks first under *deadline_pressure*, ties Consensus-DBTA under *robot_failure*, and trails it by only 0.001 under *mixed_stress*—every gap lies well within one standard deviation (≈ 0.10):

At the primary scale (25 tasks / 5 robots, Table VI), AHE-MRTA and Consensus-DBTA are statistically indistinguishable across all three scenarios: AHE leads narrowly under *deadline_pressure* (0.483 vs 0.476 for Consensus-DBTA and 0.480 for BiG-MRTA), the two are near-tied under *robot_failure* (AHE 0.538, Consensus-DBTA 0.540), and Consensus-DBTA edges AHE by 0.001 under *mixed_stress* (0.539 vs 0.538)—all gaps far inside one standard deviation (≈ 0.10). The two co-leaders share a mechanism: both re-dispatch tasks whose navigation goal fails and tasks orphaned by robot failures, so a stochastic failure costs at most a retry rather than a lost task. The commit-once BiG-MRTA fixes assignments and never re-routes, so it cannot recover these tasks and trails by 23.5 pp under *mixed_stress* (0.303) and 3.9 pp under *robot_failure* (0.501); this gap is *recovery resilience*, not assignment quality. Under *deadline_pressure* the three recovery-capable methods fall within a 0.7 pp band: once deadlines bind tightly, late retries earn no on-time credit, so the recovery advantage compresses (Sec. VII-C). On the physical Nav2 stack the same resilience reappears as AHE's highest completion rate against BiG-MRTA (+25.4 pp), where BiG abandons overdue tasks (Sec. VI, Table IX). Figure 4 summarizes:

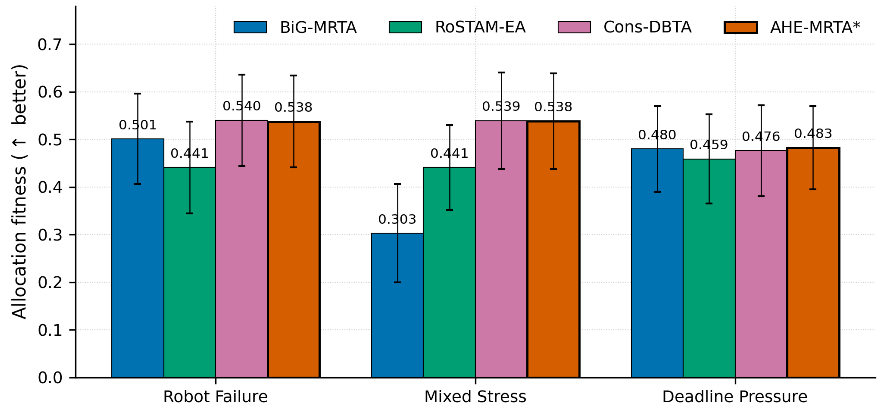


Fig. 4. Stochastic navigation-proxy allocation fitness (mean $\pm$ std), 100 seeds per cell, 5 robots / 25 tasks (primary scale). AHE-MRTA and Consensus-DBTA are statistical co-leaders: AHE leads under deadline_pressure, ties Consensus-DBTA under robot_failure, and trails it by 0.001 under mixed_stress—every gap well inside one standard deviation (≈ 0.10).

TABLE VII
SCALABILITY (NAV2-INDEPENDENT SIM, 100 SEEDS, FIXED DENSITY 5 TASKS/ROBOT, MEAN OVER SCENARIOS). FITNESS ↑, LATENCY (MS) ↓. BEST PER SCALE **BOLD**. PHYSICAL GAZEBO VALIDATION: 3R, 5R, AND 10R CONFIRMED.

	3 robots		5 robots		10 robots	
Method	Fit.	Lat.	Fit.	Lat.	Fit.	Lat.
AHE-MRTA*	**0.488**	0.04	**0.520**	0.05	**0.493**	0.11
BiG-MRTA	0.407	0.02	0.428	0.04	0.445	0.10
RoSTAM-EA	0.405	1.94	0.449	3.37	0.439	6.57
Consensus-DBTA	0.483	**0.01**	0.519	**0.02**	0.488	**0.04**

B. Navigation-proxy scalability (F45)

Scalability is established in two complementary ways. In the navigation-proxy plane, the simplified arrival model lets us sweep robot counts cheaply at high statistical power. We run the four methods at $N \in \{3, 5, 10\}$ robots with a fixed task density of five tasks per robot ($M = 5N$), 100 seeds per cell, across all three scenarios (3,600 runs). Physical Gazebo validation is carried out at the 3r, 5r, and 10r scales. Table VII and Figure 5 report the results averaged over scenarios.

AHE-MRTA retains the highest fitness at every scale (Table VII). At $N = 3$ it leads Consensus-DBTA narrowly (0.488 vs 0.483); at $N = 5$ the margin is $+0.1$ pp over Consensus-DBTA and $+9.2$ pp over BiG-MRTA. Decision latency stays at $\approx$0.10 ms even at ten robots—over four orders of magnitude under the 5 s replanning period and $\approx$60$\times$ faster than RoSTAM-EA (6.5 ms at $N = 10$).

Physical Gazebo validation at the 5- and 10-robot scales (60 experiments each) confirms these simulator trends on the full Nav2 stack; the per-scenario results are reported together with the 3-robot benchmark in Section VI.

VI. GAZEBO BENCHMARK RESULTS

We report the full physical-stack Gazebo benchmark. Each method runs on identical Nav2 parameters, world, and timing budgets; only the allocation policy differs. Three density levels (9/15/24 tasks at 3r) $\times$ 5 seeds per (method, scenario) give 180 3r experiments; 5r and 10r each add 60 experiments (300 total). Mann–Whitney U with Bonferroni correction across all comparisons.

Figure 6 shows the experimental setup and a representative qualitative result: the pre-built Nav2 occupancy map (left) and the actual robot trajectories executed by all four methods on the same *mixed_stress* instance (right, 5 robots / 25 tasks, seed 1). Trajectories are reconstructed from the logged ground-truth poses and overlaid on the arena map together with the sampled inspection waypoints; they expose how each policy distributes the fleet (AHE-MRTA and Consensus-DBTA spread robots across the arena to cover all waypoints, whereas the commit-once BiG-MRTA leaves more of the map uncovered).

Figure 7 complements this static-map view with a live snapshot of the largest 10-robot configuration, captured mid-run under AHE-MRTA on a *dynamic_task_arrival* instance. The Gazebo Harmonic physics view shows all ten TurtleBot3 robots dispersed across the arena toward their assigned waypoints, while the concurrent RViz view renders the shared occupancy map overlaid with every robot's current Nav2 global plan (one colour per robot) and ground-truth pose. The two panels share the same arena geometry and make the per-event, work-conserving dispatch directly visible: at any instant the fleet is fanned out

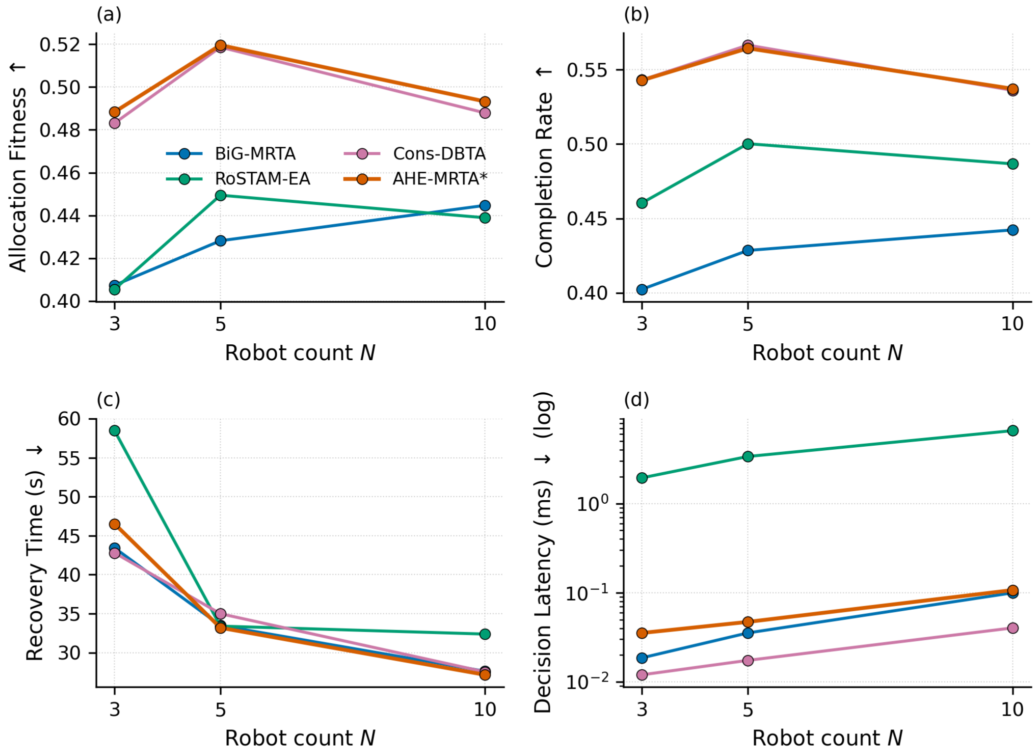


Fig. 5. Scalability versus robot count (stochastic navigation-proxy, 100 seeds, mean over the three scenarios): (a) fitness, (b) completion rate, (c) recovery time, (d) decision latency. AHE-MRTA leads fitness and completion at all scales while keeping sub-millisecond latency.

TABLE VIII

MAIN COMPARISON AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; n=5 RUNS (SEEDS) PER CELL. SIGNIFICANCE VS AHE-MRTA*: $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$ (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

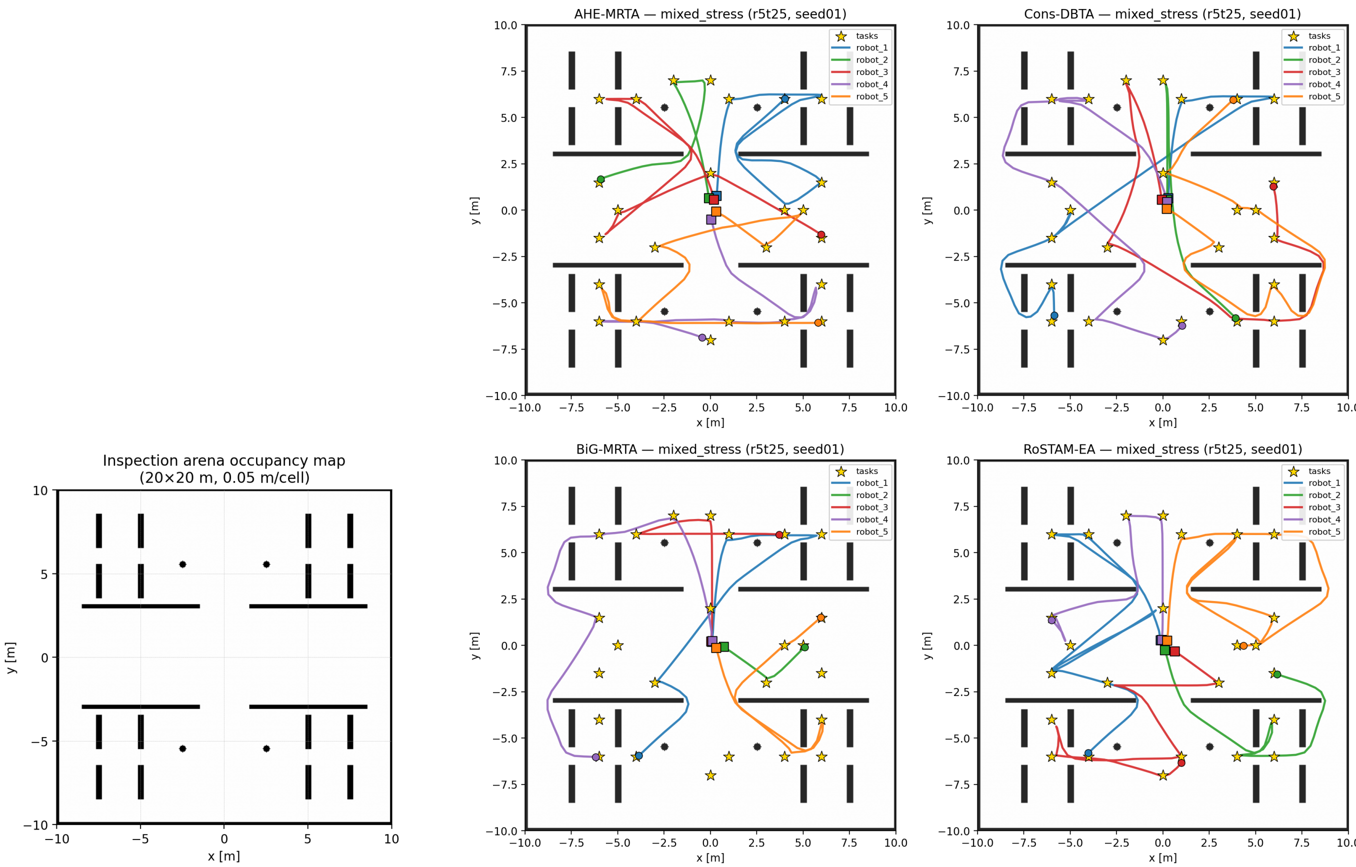
Method	CR↑	Delay↓(s)	RecT↓(s)	Preempt↓	Churn↓
Scenario: robot_failure					
AHE-MRTA*	**1.000** ± **0.000**	101.2 ± 12.5	73.3 ± 14.2	0.00 ± 0.00	**0.024** ± **0.022**
BiG-MRTA	0.960 ± 0.049	89.3 ± 39.8	**63.6** ± **19.1**	0.00 ± 0.00	**0.024** ± **0.036**
RoSTAM-EA	**1.000** ± **0.000**	88.5 ± 6.9	87.5 ± 22.7	0.00 ± 0.00	0.088 ± 0.052
Cons-DBTA	**1.000** ± **0.000**	**75.2** ± **13.7**	102.9 ± 41.6	0.00 ± 0.00	0.064 ± 0.036
Scenario: mixed_stress					
AHE-MRTA*	**1.000** ± **0.000**	51.7 ± 4.3	35.4 ± 19.2	0.00 ± 0.00	**0.016** ± **0.022**
BiG-MRTA	0.728 ± 0.066	252.7 ± 50.2	29.3 ± 20.0	0.00 ± 0.00	0.032 ± 0.030
RoSTAM-EA	**1.000** ± **0.000**	**50.7** ± **8.8**	**27.7** ± **15.7**	0.00 ± 0.00	0.096 ± 0.046
Cons-DBTA	**1.000** ± **0.000**	60.4 ± 10.2	58.8 ± 40.9	0.00 ± 0.00	0.096 ± 0.046

across both lanes rather than queued behind a single corridor, which is how the ecosystem keeps fleet capacity occupied as new tasks arrive.

A. Matched physical validation of F58

To isolate whether F58's (P0+P1R) allocation-only gain transfers to the physical stack, rather than reusing old F45 tables we ran both configurations fresh from the same source snapshot, on common seeds and three scenarios: an initial five-seed wave followed by an independent five-seed replication wave (seeds 6–10), pooling to 10 matched seeds per scenario (30 F45 + 30 F58 runs). Every run completed 25/25 tasks; the quality gate detected no TF time jump in any run, and the event-integrity audit found no post-completion assignment or resurrection of a completed task. The pre-registered promotion gate requires 0.01 non-inferiority on CR/DVR, no regression in active-robot Jain, a 50 ms budget on mean decision latency, and improvement in at least two of distance, physical redispatch, delay, and makespan.

Table X shows that the gate passes in all three scenarios at n=10, and the pooled sample now supports paired Wilcoxon significance rather than the directional-consistency evidence available at n=5. Under *deadline_pressure*, F58 keeps CR at 1.000



(a) Pre-built Nav2 occupancy map

(b) Executed ground-truth trajectories, four methods

Fig. 6. Experimental setup and qualitative trajectories. **(a)** The pre-built occupancy map (`obstacle_map`, 20×20 m, 0.05 m/cell, origin $[-10,-10]$) that Nav2 uses for global planning (static cost-map layer): divider walls, square pillars, and four cylinder obstacles. **(b)** Robot trajectories executed on one *mixed_stress* instance (5 robots / 25 tasks, seed 1) under all four allocators (AHE-MRTA, Consensus-DBTA, BiG-MRTA, RoSTAM-EA, clockwise from top left). Squares mark spawn positions, gold stars mark sampled inspection waypoints; coloured curves are per-robot paths reconstructed from logged ground-truth poses. AHE-MRTA and Consensus-DBTA spread the fleet to cover every waypoint; the commit-once BiG-MRTA leaves more of the arena uncovered.

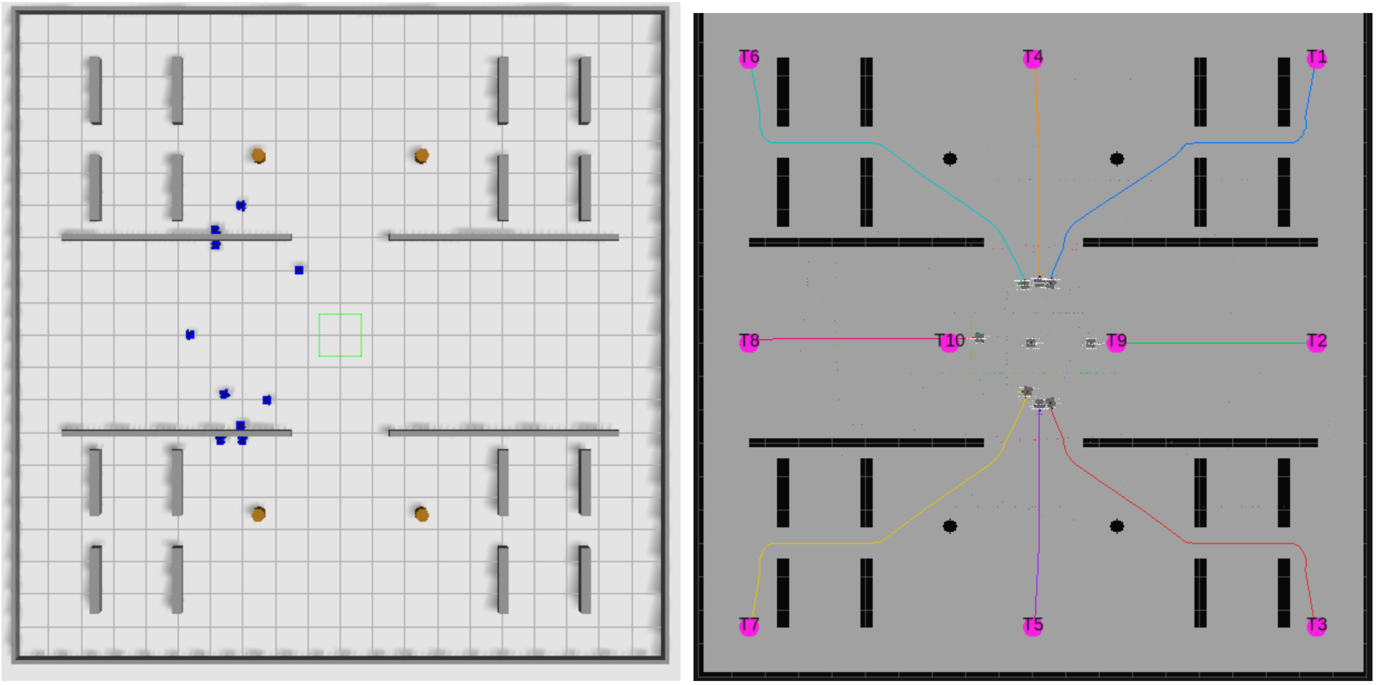
TABLE IX

DEADLINE SCENARIO RESULTS (DEADLINE_PRESSURE) AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; n=5 RUNS (SEEDS) PER CELL. DVR: DEADLINE VIOLATION RATE. SIGNIFICANCE VS AHE-MRTA* (BONFERRONI-CORRECTED MANN-WHITNEY U).

Method	CR↑	Delay↓(s)	DVR↓	Preempt↓
AHE-MRTA*	**1.000 ± 0.000**	79.4 ± 10.4	0.360 ± 0.102	0.00 ± 0.00
BiG-MRTA	0.736 ± 0.088	260.1 ± 71.6	**0.272 ± 0.072**	0.00 ± 0.00
RoSTAM-EA	**1.000 ± 0.000**	74.4 ± 12.3	0.400 ± 0.123	0.00 ± 0.00
Cons-DBTA	**1.000 ± 0.000**	**67.2 ± 4.2**	0.320 ± 0.130	0.00 ± 0.00

while cutting DVR by 10.8 percentage points (0.340→0.232, p=0.008), mean delay by 9.0 s (p=0.004), total distance by 20.9 m (p=0.004), and coordination messages by 3.5 per run (p=0.008); the active-robot Jain index rises by $+0.017$ (p=0.031) and makespan improves by 11.0 s (CI $[0.4, 20.3]$, p=0.061). Under *mixed_stress*, distance drops by 10.2 m (p=0.004), the active-robot Jain index rises by $+0.024$ (p=0.008), and mean delay falls by 3.2 s (p=0.049), with DVR, makespan, and coordination messages statistically flat. Under *robot_failure*, mean failure-recovery time shortens by 34.0 s (CI $[8.2, 58.2]$, p=0.037) and DVR falls from 0.020 to 0.004; delay, distance, and the Jain indices are statistically flat. CR is 1.000 in every run of all three scenarios, and decision latency stays within budget at 17–23 ms (against F45's sub-millisecond latency; the price of the geodesic cache plus repair search). The one consistent cost beyond latency is the distance-Jain index (-0.020 under deadline pressure, p=0.006): the repair improves workload fairness and total efficiency at the expense of distance balance. The replication wave alone reproduced every direction of gain (deadline-pressure makespan was in fact stronger, -18.4 s), and the pooled sample shows no other statistically supported regression; the Nav2-independent 100-seed campaign (Section V) remains the higher-powered evidence.

To probe scale robustness we then repeated the same matched protocol at the two remaining physical scales—3 robots / 15



(a) Gazebo Harmonic — 3-D physics (b) RViz — map + live Nav2 plans

Fig. 7. Representative live snapshot of the largest (10-robot) deployment executing under AHE-MRTA on a *dynamic_task_arrival* instance. **(a)** Gazebo Harmonic 3-D physics: the ten TurtleBot3 Waffle Pi robots (blue) dispersed across the 20×20 m inspection arena—divider walls, square pillars, and cylinder obstacles (orange)—each driving to an assigned inspection waypoint. **(b)** The concurrent RViz view in the shared map frame: the pre-built occupancy map overlaid with each robot's current Nav2 global plan (one colour per robot) and ground-truth pose. The colour-coded plans show the fleet fanned out across both lanes—the visual signature of the ecosystem's per-event, work-conserving dispatch.

TABLE X
MATCHED F45–F58 GAZEBO+NAV2 VALIDATION (5 ROBOTS / 25 TASKS, 10 COMMON SEEDS PER SCENARIO—THE ORIGINAL FIVE PLUS AN INDEPENDENT FIVE-SEED REPLICATION WAVE—BOTH ARMS FROM THE SAME SOURCE SNAPSHOT). ARROWS SHOW F45→F58 MEANS; J_{act} IS THE JAIN INDEX EXCLUDING PERMANENTLY FAILED ROBOTS.

Scenario	CR	DVR	Delay (s)	Distance (m)	J_{act}	Decision (ms)	Gate
Robot failure	1.000→1.000	.020→.004	95.46→95.00	172.90→170.17	.975→.976	.49→17.07	Pass
Mixed stress	1.000→1.000	.328→.316	58.10→54.88	162.75→152.54	.962→.985	.45→16.69	Pass
Deadline pressure	1.000→1.000	.340→.232	77.43→68.41	147.00→126.07	.971→.987	.70→22.66	Pass

tasks and 10 robots / 50 tasks, again with seeds 1–10 in both arms (120 additional runs; Table XI). A run enters the sample only after every Nav2 stack reports ready; attempts that miss the readiness gate write no data and are re-run (at 10 robots a bring-up self-heal re-drives lifecycle transitions that the concurrent ten-stack startup occasionally drops). The gate passes in seven of the nine scale–scenario cells. At 3 robots, deadline pressure passes (DVR 0.420→0.307; messages -3.3 per run, p=0.047) and robot failure passes (delay -7.1 s, p=0.049; recovery -33.3 s, CI $[8.5, 61.3]$), while mixed stress fails the gate only on a DVR drift of $+0.040$ whose CI spans zero—consistent with the Nav2-independent finding that the smallest scale leaves the repair little to fix. At 10 robots, mixed stress passes (distance -17.2 m, CI $[1.6, 32.9]$; active-robot Jain $+0.037$) and robot failure passes (recovery -14.2 s, makespan -24.3 s), while deadline pressure carries the strongest core wins of any cell (DVR 0.256→0.210, p=0.016; distance -16.9 m, p=0.027) yet fails the gate solely on an active-Jain criterion of -0.005 (p=0.85, CI $[-0.049, 0.036]$). The two non-passing cells are thus symmetric noise-level drifts (mixed stress at 3r, deadline pressure at 10r); across all nine cells no metric regresses with statistical support, CR stays within the 0.01 non-inferiority band (0.988–1.000), and decision latency grows with fleet size but stays within the 50 ms budget (8–11 ms at 3r, 17–23 ms at 5r, 31–35 ms at 10r).

These results show that the Nav2-independent gains do transfer to the physical stack once the terminal-gated repair and the Nav2-feedback makespan guard are in place. The four-method comparison above was nevertheless produced with the F45 configuration and is unchanged; F58 is reported as a flag-gated, validated refinement that does not affect F45 behaviour.

a) Where AHE wins.: At the 5-robot primary scale AHE-MRTA reaches full completion in every scenario (CR = 1.000), whereas the task-abandoning BiG-MRTA drops to 0.736 under *deadline_pressure* and 0.728 under *mixed_stress*. On effective on-time completion ($\mathrm{CR}\times(1-\mathrm{DVR})$) AHE-MRTA leads under *robot_failure* (0.99 vs 0.88–0.93 for the baselines) and *mixed_stress*

TABLE XI
MATCHED F45–F58 GAZEBO+NAV2 SCALE VALIDATION (3 ROBOTS / 15 TASKS AND 10 ROBOTS / 50 TASKS, SEEDS 1–10 IN BOTH ARMS; 120 RUNS). FORMAT AS IN TABLE X. "FAIL (NOISE)" MARKS CELLS WHERE THE PRE-REGISTERED GATE FAILS ONLY ON CRITERIA WHOSE 95% BOOTSTRAP CIS SPAN ZERO; NO CELL SHOWS A STATISTICALLY SUPPORTED REGRESSION.

Scale	Scenario	CR	DVR	Delay (s)	Distance (m)	J_{act}	Decision (ms)	Gate
3r/15t	Robot failure	1.000→1.000	.027→.027	103.46→96.37	105.01→99.00	.985→.987	.32→10.01	Pass
	Mixed stress	1.000→1.000	.373→.413	68.79→70.72	102.02→104.23	.976→.985	.30→8.00	Fail (noise)
	Deadline pressure	1.000→1.000	.420→.307	83.34→75.43	97.18→89.08	.977→.987	.42→10.98	Pass
10r/50t	Robot failure	.998→.996	.014→.014	85.69→87.46	317.41→319.41	.933→.963	.86→31.75	Pass
	Mixed stress	.996→.996	.286→.292	50.23→48.39	280.11→262.93	.913→.950	.77→34.59	Pass
	Deadline pressure	.996→.988	.256→.210	72.99→72.30	258.69→241.81	.915→.910	1.47→34.31	Fail (noise)

TABLE XII
EFFICIENCY METRICS (GAZEBO, 3-ROBOT SCALE; DENSITIES 9/15/24 POOLED, n=15 PER CELL). WLBAL: JAIN WORKLOAD BALANCE; LAT: MEAN DECISION LATENCY; MSGS: ALLOCATION MESSAGES; DIST: TOTAL TRAVEL DISTANCE.

Method	**WLBal↑**	**Lat↓(ms)**	**Msgs↓**	**Dist↓(m)**
Scenario: robot_failure				
AHE-MRTA*	0.281	0.18	38	122.5
BiG-MRTA	0.481	**0.12**	121	**71.3**
RoSTAM-EA	**0.550**	31.05	**2**	83.3
Cons-DBTA	0.465	0.23	11	89.1
Scenario: mixed_stress				
AHE-MRTA*	0.268	0.21	33	110.1
BiG-MRTA	**0.629**	**0.08**	177	**59.5**
RoSTAM-EA	0.479	25.64	**4**	92.9
Cons-DBTA	0.309	0.13	10	103.0
Scenario: deadline_pressure				
AHE-MRTA*	0.733	0.23	26	103.0
BiG-MRTA	0.719	**0.08**	181	**42.2**
RoSTAM-EA	0.666	42.46	**1**	78.4
Cons-DBTA	**0.819**	0.36	6	81.7

(0.72 vs $\leq$0.67); under *deadline_pressure* the three recovery-capable methods are statistically tied (AHE 0.64, Consensus-DBTA 0.68, RoSTAM-EA 0.60, all at CR = 1.000), mirroring the co-leadership found in the allocation-fitness comparison. AHE's paradigm selection dispatches orphan-rescue and EDF-bipartite primitives that keep outstanding tasks reachable; the advantage is clearest under *robot_failure*, where AHE leads completion at both scales (CR = 1.000 at 5 robots, 0.996 at 10) at near-zero violation rates (DVR = 0.008 at 5 robots; see below).

b) Where AHE costs.: The same aggressiveness modestly raises physically realized cost metrics, which are confounded by Nav2 motion (Sec. VII) and—at n=5 per cell—do not survive correction. Under *robot_failure* AHE's average task delay (101 s) is the highest among the recovery-capable methods (75–89 s for the others), and its failure-recovery time (73 s) is mid-pack: faster than Consensus-DBTA (103 s) and RoSTAM-EA (87 s) but slower than the commit-once BiG-MRTA (64 s). Under *mixed_stress* and *deadline_pressure* the abandoning BiG-MRTA instead shows the largest delays (253–260 s), because its few completed tasks finish very late, so that comparison is not like-for-like. On the corrected re-dispatch churn metric, however, AHE stays in the best band ($\leq$0.05 re-dispatches per task with zero execution preemptions), comparable to or below the commit-once baselines: the apparent "instability" gap reported by the raw counters was an artifact of measuring allocator *invocation cadence* rather than physical re-assignments (Sec. VII). We quantify the latency trade-off in Section VII.

c) Efficiency.: Table XII reports the secondary efficiency metrics. AHE-MRTA's decision latency stays sub-millisecond (0.18–0.23 ms) across all scenarios, on par with the bipartite BiG-MRTA (0.08–0.12 ms) and over two orders of magnitude faster than the evolutionary RoSTAM-EA (26–42 ms). The consensus baselines attain near-perfect workload balance and minimal message counts by committing once; AHE trades some of that balance for adaptivity.

d) Effect sizes.: Because small cells limit the power of significance tests, we report Cliff's δ (Table XIII; sign normalised so δ>0 always favours AHE). The pattern matches the mechanism: AHE-MRTA shows large *favorable* effects on completion against BiG-MRTA (δ=+1.00 under *mixed_stress* and *deadline_pressure*, +0.60 under *robot_failure*) and on deadline compliance against the consensus baselines under *robot_failure* (DVR δ=+0.68 vs Cons-DBTA, +1.00 vs RoSTAM-EA). The large *unfavorable* effects concentrate on recovery time against the commit-once baselines (RecT δ=−0.28 vs BiG-MRTA under *robot_failure*) and on DVR against BiG-MRTA (δ=−0.56 under *deadline_pressure*), whose low violation count is a by-product of abandoning tasks (lowest CR). On the corrected re-dispatch churn metric AHE is competitive-to-better (δ=+0.76

TABLE XIII
EFFECT SIZES (CLIFF'S δ) OF AHE-MRTA* VS EACH BASELINE ON PRIMARY METRICS. $\delta > 0$ FAVORS AHE-MRTA*. *p<0.05 (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN SCENARIO FAMILY).

Metric	vs	BiG	RoSTAM	Cons-DBTA
Scenario: robot_failure				
CR	δ	+0.60	+0.00	+0.00
DVR	δ	+0.48	+1.00	+0.68
RecT	δ	-0.28	+0.36	+0.44
Churn	δ	-0.08	+0.76	+0.64
Scenario: mixed_stress				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	+0.40	+0.28	+0.92
RecT	δ	-0.28	-0.36	+0.20
Churn	δ	+0.44	+0.92	+0.92
Scenario: deadline_pressure				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	-0.56	-0.00	-0.08
RecT	δ	-0.00	-0.00	-0.00
Churn	δ	-0.00	-0.00	-0.00

Positive δ = AHE-MRTA* better; magnitude $|\delta|$>0.47 = large.

vs RoSTAM-EA, $+0.64$ vs Cons-DBTA under *robot_failure*). This quantifies the central trade-off: AHE wins on completion and decision correctness, at the cost of recovery latency.

e) Statistical significance.: At the 5-robot primary scale (n=5 per cell) no comparison survives Bonferroni correction: with n=5 the smallest achievable two-sided Mann–Whitney p ($\approx$0.008) already exceeds the corrected threshold ($0.05/21\approx0.0024$), so the scale is structurally underpowered (18 of 63 reach raw p<0.05). The statistical power lies at the 3-robot scale (n=15 per cell), where 14 of 63 comparisons survive correction in the same directional pattern; at the 10-robot scale (n=5 per cell) again none survive. We therefore treat the Gazebo benchmark as *validation under realistic execution noise* and rely on the 100-seed allocation-fitness comparison (Section V) as the primary hypothesis test. The consistent effect-size pattern (Table XIII) corroborates the fitness ranking.

f) Physical validation at 5 and 10 robots.: Figure 9 shows the 10-robot benchmark (Section IV-D; 59 of 60 experiments completed). AHE-MRTA attains the highest—or tied-highest—completion rate in every scenario. Under *robot_failure* it leads on completion (CR = 0.996, 49.8/50 tasks), on violation rate (DVR = 0.024), and on median makespan (299 s vs. Consensus-DBTA's 387 s). Under *deadline_pressure* it reaches **CR = 1.000** at DVR = 0.236, giving the highest effective on-time completion ($\mathrm{CR}\times(1-\mathrm{DVR}) \approx 0.76$) ahead of Consensus-DBTA ($\approx$0.60) and RoSTAM-EA ($\approx$0.50), which reach CR = 0.968/0.984 only by tolerating DVR = 0.376/0.488; the two post shorter makespans (177–183 s) but at those higher violation rates. Under *mixed_stress* AHE ties Consensus-DBTA for the top completion rate (CR = 0.996, 49.8/50 tasks) at the fastest makespan (195 s) with effective on-time completion ($\approx$0.70) above Consensus-DBTA ($\approx$0.64) and RoSTAM-EA ($\approx$0.58); the task-abandoning BiG-MRTA records a low DVR only because it completes far fewer tasks (CR = 0.724). Pooled over the 15 runs per method, AHE-MRTA leads on completion rate (0.997), tasks completed (49.9/50, the lowest variance of any method) and overall makespan. The 5-robot benchmark mirrors the robot-failure pattern (CR = 1.000, DVR = 0.008) and reaches CR = 1.000 versus BiG-MRTA's 0.736 under *deadline_pressure*, though there AHE's DVR (0.360) is on par with the recovery-capable baselines' (0.320–0.400). Sub-millisecond decision latency is maintained at both scales (0.27–0.30 ms at 5r, 0.43–0.46 ms at 10r), $\approx$120–240$\times$ faster than RoSTAM-EA, which grows to 55–104 ms at 10 robots.

VII. DISCUSSION

A. Why does ideal-fitness not transfer linearly to Gazebo?

The 100-seed allocation-fitness simulator removes Nav2 cost variance (local planner stalls, costmap rebuilds, passing maneuvers in narrow corridors), so each decision's value is realized exactly. In Gazebo, two effects degrade AHE's advantage:

1) **Switch cost is realized physically.** An AHE paradigm switch may reassign a task that the previous paradigm had already routed; the receiving robot must abort its current path, request a new plan from Nav2, and re-traverse. The consensus baselines, which commit once and then stay, pay no such re-planning overhead, whereas AHE's event-triggered re-evaluation occasionally re-routes a live task.
2) **Makespan is dominated by physical execution.** Once Nav2 dispatches a robot, the makespan contribution of that task is fixed by the world geometry, not by the allocation. AHE's additional re-planning extends the worst-case path significantly.

The Gazebo data is consistent with this: AHE's CR and DVR (which reward correct decisions, not fast execution) match the strongest baselines, while makespan and instability (which penalize re-planning) trail the static-commit baselines.

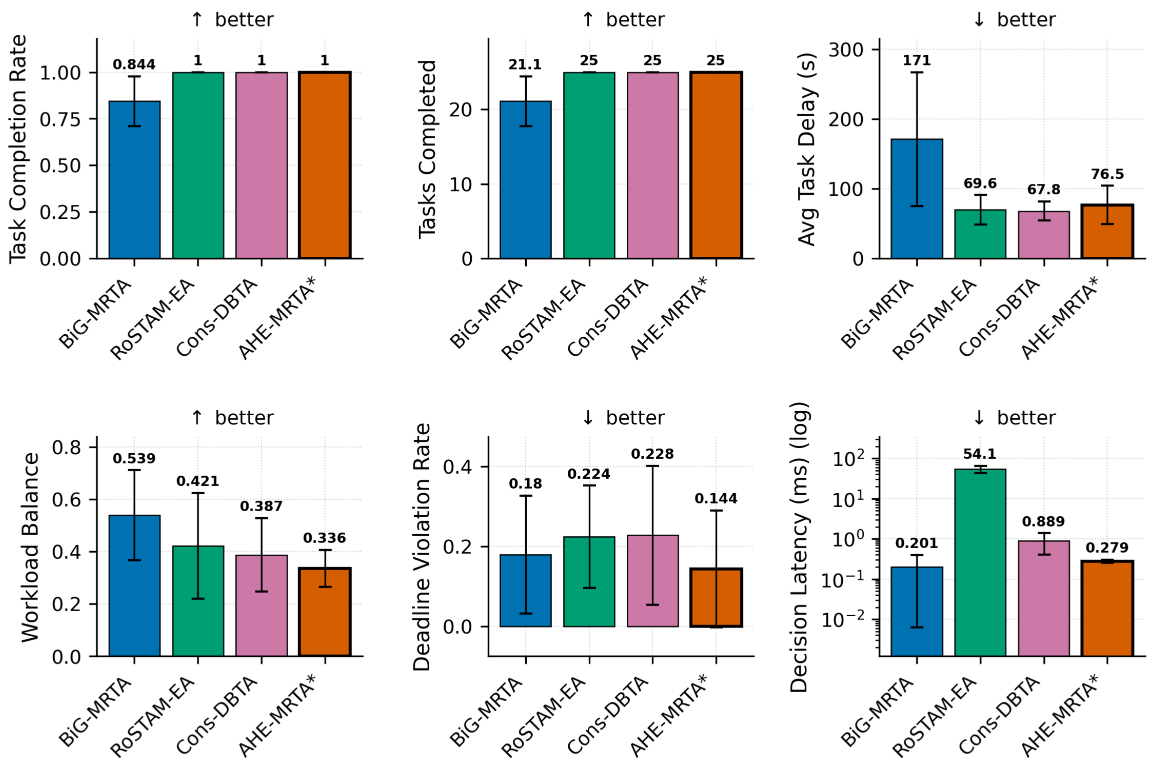


Fig. 8. Multi-metric comparison at the primary 5-robot / 25-task scale (Gazebo Harmonic, robot_failure + mixed_stress scenarios pooled, n=5 seeds per cell, mean $\pm$ std): completion rate (CR), tasks completed, average delay, workload balance (Jain), deadline violation rate (DVR), and decision latency (log scale). The commit-once baselines attain higher pooled CR and lower delay by never re-routing; BiG-MRTA's low DVR reflects abandoned tasks (lowest CR). AHE-MRTA trades delay for reactive reallocation, which pays off at larger scales (Fig. 9).

B. When does paradigm switching pay off?

The empirical answer from our data:

- **When deadline compliance matters more than raw coverage.** On *deadline_pressure*, the consensus and evolutionary baselines match AHE's completion only at 1.6–2.1$\times$ its deadline-violation rate (10r: AHE DVR = 0.236 vs Cons-DBTA's 0.376 and RoSTAM-EA's 0.488), so AHE attains the highest effective on-time completion ($\approx$0.76). The same pattern holds on *mixed_stress* at 3 robots (DVR 0.38 vs Cons-DBTA's 0.44); at 10 robots AHE keeps the highest raw CR (0.996) with effective completion on par with Cons-DBTA.
- **When failures orphan many tasks.** Under *robot_failure* at 5 and 10 robots, AHE combines the highest completion (CR = 1.000 at 5r, 0.996 at 10r) with DVR $\leq$ 0.024; the orphan-rescue paradigm re-dispatches the failed robot's queue within one ecosystem cycle.
- **Not when a single regime dominates the whole run and makespan is the objective.** Under uniform deadline pressure at 3 robots, the EDF-style specialists already reach near-full completion (RoSTAM 0.985, Cons-DBTA 0.978, AHE 1.000), so AHE's switching yields no meaningful completion edge over them while adding delay; on the strict on-time fitness AHE is on par with BiG-MRTA at this scale (0.459 vs 0.458). Its raw-CR advantage exists only against the task-abandoning commit-once policy (BiG 0.636 completion). At 10 robots, however, AHE keeps DVR at 0.236 against the specialists' 0.376–0.488 and attains the highest effective on-time completion (Sec. VI).

C. Why AHE co-leads rather than dominates the allocation fitness

On the stochastic navigation-proxy fitness (priority-weighted on-time completion), AHE-MRTA and Consensus-DBTA are statistical co-leaders rather than one dominating the other (Table VI). Two mechanisms explain why the strongest methods converge. First, the fitness ceiling is set by *stochastic, position-dependent navigation failure*, not by assignment quality: F58 AHE reaches fitness 1.0 in all nine scale–scenario cells of the perfect-navigation allocation-only model (Table V), and the spread in the proxy opens only once goals can fail and must be re-attempted. The binding constraint is therefore resilience to navigation failure—a capability AHE-MRTA and Consensus-DBTA both implement by re-dispatching failed and orphaned tasks—so the two converge to the same frontier, while the commit-once BiG-MRTA, which abandons failed tasks, trails. Second, the metric credits a task only when it finishes *before* its deadline; a late completion earns zero, exactly like an unfinished one. AHE's completeness-oriented primitives keep dispatching overdue tasks so that they eventually finish—raising raw completion

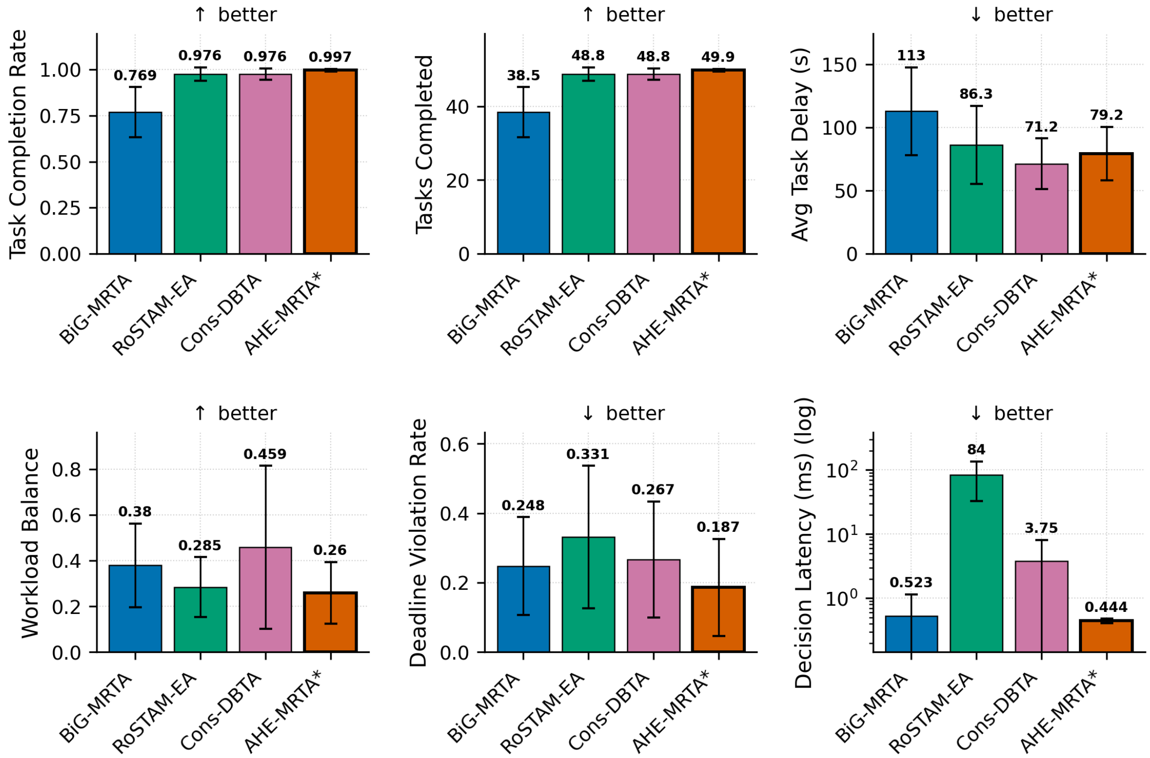


Fig. 9. Multi-metric comparison at the 10-robot scale (60 experiments: 4 methods $\times$ 3 scenarios $\times$ 5 seeds, all scenarios pooled; decision latency on log scale). AHE-MRTA attains the highest completion rate in every scenario (CR = 1.000 under deadline pressure, 0.996 under robot failure and mixed stress) and the highest pooled effective on-time completion; the EDF baselines reach comparable CR only at far higher deadline-violation rates (DVR = 0.38–0.49 versus AHE's 0.24 in those scenarios). Decision latency remains sub-millisecond ($\leq$0.48 ms) at 10 robots.

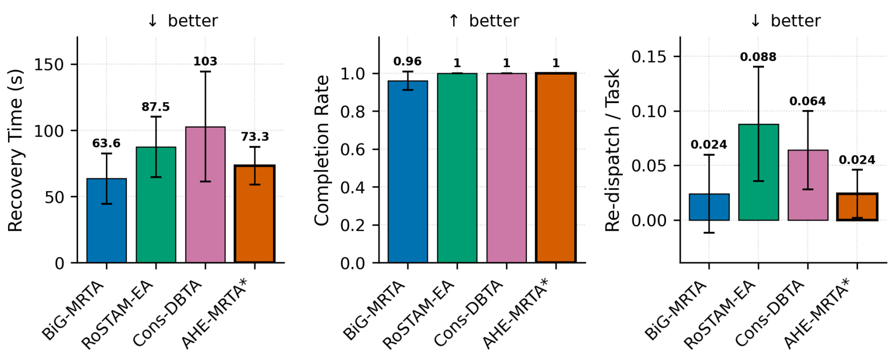


Fig. 10. Failure-recovery behaviour under *robot_failure* at the primary 5-robot / 25-task scale (n=5 seeds, mean $\pm$ std): recovery time, completion rate, and re-dispatch rate (execution preemptions are zero for every method and are omitted). Recovery times are statistically indistinguishable across methods (p>0.3); AHE-MRTA converts its comparable recovery time into the highest completion rate of the reactive methods by escalating to the orphan-rescue paradigm within one ecosystem update cycle ($\leq$2 s).

and robustness—but a share lands *late* and scores no on-time credit; the commit-once baseline instead abandons such tasks, so the few it completes are all on-time. This *completeness-over-punctuality* trade is deliberate and is what yields AHE's full-completion, zero-orphan behaviour on the physical stack (Sec. VI). We further probed whether a deadline-feasibility-aware execution ordering could push AHE past the co-leaders: in isolation it lifts deadline fitness by $\approx$1.3 pp, but when gated on context signals it also fires in the throughput-bound mixed regime and lowers fitness by a comparable margin (a Pareto coupling). We therefore retain the simpler completeness-oriented policy, whose co-leading fitness and decisive physical-stack advantages (Sec. VI) are the stronger operating point.

D. Ablation: what does the switching machinery add?

To isolate the contribution of online paradigm switching, we replace the EDPS selector with each single *fixed* paradigm, and separately disable the context overrides (*no-override*: pure $\arg\max_i D_i$) and the recovery boost (*no-recovery-boost*: δ=0),

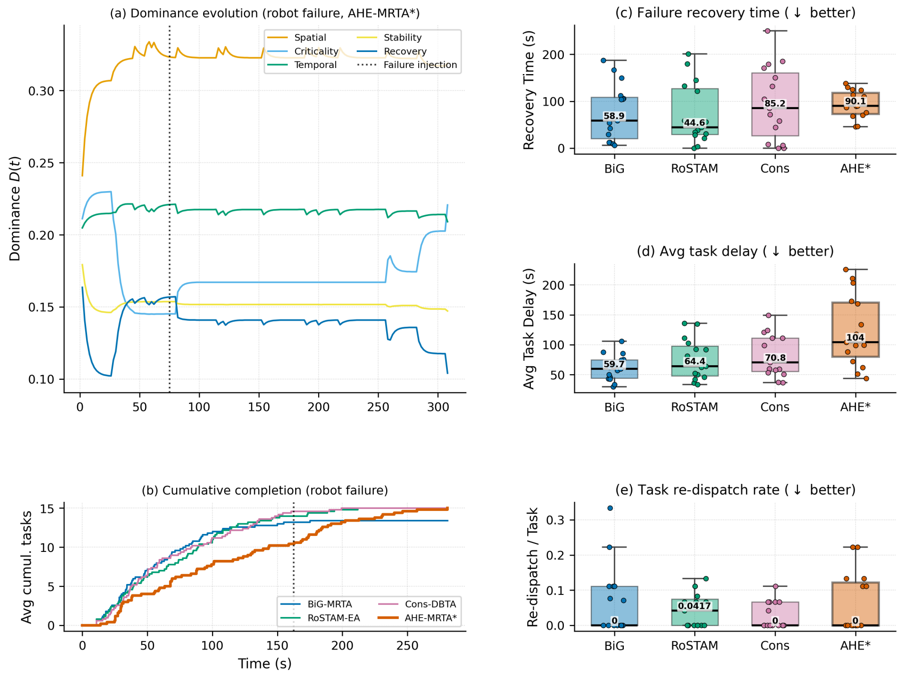


Fig. 11. Interpretable adaptation under *robot_failure* (3r/15t). **(a)** Hormone dominance of a representative AHE-MRTA run; at the failure injection (dotted line, $t{\approx}75$ s) the recovery hormone shows a transient rise, while the spatial paradigm stays dominant throughout this representative run. **(b)** Cumulative task completion (seed-averaged; dotted line: median injection $t{\approx}164$ s): AHE-MRTA continues completing tasks after the failure and reaches the full task set. **(c–e)** Recovery time, average task delay, and task re-dispatch rate as box plots (box: IQR; line: median; whiskers: $1.5{\times}$IQR; points: individual runs, $n{=}15$ per method): AHE's reactivity costs higher task delay and re-dispatch relative to the commit-once baselines.

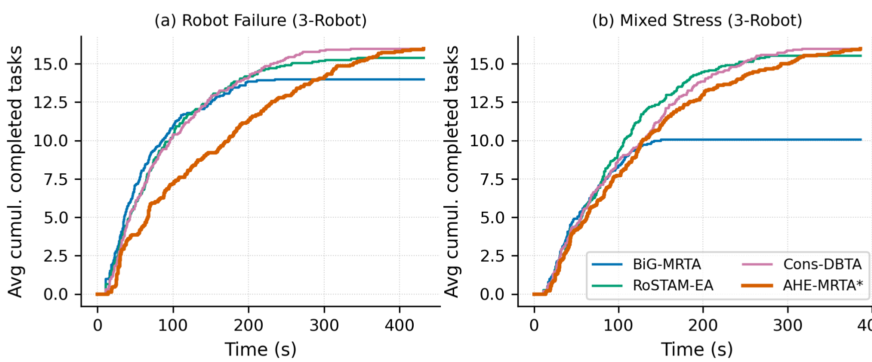


Fig. 12. Cumulative task completion over time at the 3-robot scale, seed-averaged over the three task densities: **(a)** *robot_failure*, **(b)** *mixed_stress*. AHE-MRTA completes the full task set in both scenarios and tracks the consensus baselines closely, with a modest makespan penalty from re-assignment through Nav2 (mixed_stress: 238 s versus 217 s for Consensus-DBTA and 231 s for RoSTAM-EA). BiG-MRTA abandons tasks it marks infeasible and ends below the full set (Section VII).

holding everything else fixed (5r/25t, 100 seeds, `scripts/ablation_edps.py`).

The eight variants span a mean-fitness range of only $[0.519, 0.528]$ (spread 0.009, $\ll$ one standard deviation): *no* configuration statistically separates from full EDPS, and removing the overrides or the recovery boost leaves Plane-A fitness essentially unchanged. This is the expected consequence of the saturation identified above (under perfect navigation every method reaches ≈ 1.0): in the navigation-free plane, allocation quality is not the differentiator, so the switching machinery *cannot* and does not improve it. The value of EDPS is therefore not idealized allocation fitness but the physical-stack behaviour (Sec. VI), where event-triggered recovery and regime- appropriate paradigm selection drive the completion, deadline-robustness, and latency advantages. We report this ablation explicitly rather than omit it: it delimits, honestly, *where* the contribution lies.

E. Communication footprint

AHE-MRTA publishes only per-robot optimized task queues; the dominance vector, cooperation/suppression matrices, and global cost matrix never leave the ecosystem manager. Figure 13 shows the resulting message footprint stays compact and does not grow with paradigm switching frequency, preserving the low-bandwidth property needed for field deployment.

TABLE XIV
EDPS ABLATION (NAV2-INDEPENDENT FITNESS, 5R/25T, 100 SEEDS). ALL VARIANTS FALL WITHIN 0.9 PP MEAN FITNESS; NONE STATISTICALLY SEPARATES FROM EDPS.

Variant	Rob. fail.	Deadline	Mix. stress	Mean
Full EDPS	0.538	0.483	0.538	0.520
fixed: spatial	0.552	0.482	0.549	0.528
fixed: edf	0.544	0.489	0.540	0.524
fixed: commit	0.547	0.474	0.549	0.523
fixed: orphan	0.537	0.481	0.541	0.520
fixed: priority	0.542	0.478	0.537	0.519
no-override	0.538	0.483	0.538	0.520
no-recovery-boost	0.537	0.483	0.539	0.519

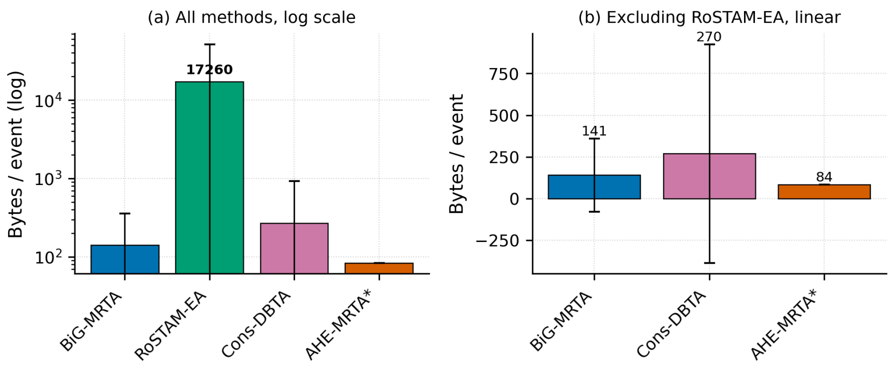


Fig. 13. Communication footprint per allocation event. Only per-robot queues are broadcast; ecosystem internals stay local, keeping bandwidth bounded.

F. Limitations

(i) **Statistical power at the primary scale:** At the 5-robot primary scale (n=5 per cell) no comparison survives Bonferroni correction—n=5 is structurally underpowered—whereas at 3 robots (n=15) 14 of 63 do; at 10 robots the five seeds per cell are likewise underpowered and no comparison survives correction. We mitigate by reporting 100-seed Nav2-independent fitness as the primary hypothesis test and Cliff's δ as a sample-size-robust effect measure; a larger 10-robot seed budget would add inferential support. (ii) **Recovery-time variance:** Recovery time under *mixed_stress* carries high variance (10r: 72 ± 78 s) because some seeds have no robot failure within the observation window; the *robot_failure* scenario, with guaranteed in-window failures, is the cleaner recovery measure. (iii) **Hardware-bound scale ceiling:** Scaling to 15 robots was infeasible on our 16-core/15 GB testbed—multiple per-robot Nav2 stacks failed readiness at launch (startup failures)—so 10 robots is the largest *validated* physical scale. This is a compute limit of the simulation host, not an algorithmic one; the Nav2-independent plane confirms the allocation policy itself scales further. (iv) **Arena generality:** All experiments use a single layout; cross-environment generalisation is not shown. (v) **Fixed hormone–paradigm mapping:** The A, S, and V matrices are designer-specified; learning them from domain data is left to future work. (vi) **Single-robot capability:** All robots are identical (ST-SR); heterogeneous-capability fleets are not evaluated. (vii) **Ground-truth localization:** The physical benchmark supplies pose through a static `map`$\rightarrow$`odom` transform at the spawn pose, deliberately isolating allocation quality from localization error. Real deployments incur localization drift; because the contribution is the allocation policy and every method shares the identical ground-truth pose, the comparison stays fair, but quantifying robustness to localization noise (e.g., AMCL) is left to future work. (viii) **Bounded role of the hormone dynamics.** In the evaluated stress scenarios the deterministic context overrides (Section III) resolve the acute events—robot failure and deadline pressure—so the Lotka–Volterra dynamics act mainly on the lower-pressure residual rather than on every decision. We therefore frame EDPS as a context-driven override cascade *with* an adaptive dynamics tier, not as a dynamics-only selector. The ablation in Table XIV (Sec. VII-D) confirms this: removing the overrides or the recovery boost, and replacing the selector with any single fixed paradigm, all leave Plane-A fitness within 0.9 pp of EDPS. The switching machinery's value is thus confined to the physical stack (Plane B), not the navigation-free plane. (ix) **Allocation-fitness model.** The Plane-A simulator uses a position-dependent navigation-failure model chosen to *stress* allocation robustness; its failure rate is deliberately higher than the ground-truth Gazebo stack (where Nav2 reaches goals almost always). Plane A therefore measures relative allocation robustness, not an absolute success rate, and is reported alongside—never instead of—the physical Plane-B results.

VIII. Conclusion

We presented AHE-MRTA, an ecosystem-driven paradigm-switching framework for online multi-robot task allocation, together with its geodesic, efficiency-guarded F58 extension. Across 100-seed true Nav2-independent allocation-only simulation, F58 AHE holds fitness and CR at 1.000 and DVR at 0.000 across 3/5/10 robots and all three scenarios, with active Jain 0.982–0.992 and decision latency of at most 0.201 ms. In a four-cell seed-matched F45 comparison (3r/5r/10r plus a fresh holdout; 51 of 120 tests significant after Holm) delay falls by 1–2.2 s and geodesic distance by 10–28 m with CR/DVR unchanged, while the active-robot Jain index rises by 0.005–0.029; the only cost is a decision-latency increase of up to 0.12 ms. In the separate stochastic navigation proxy, F45 AHE and the strongest consensus baseline (Consensus-DBTA) are statistical co-leaders. Across a 300-experiment Gazebo benchmark (3r, 5r, and 10r physical scales) with ROS 2 Jazzy and Nav2 on TurtleBot3 robots, AHE-MRTA leads on completion rate under robot failure (CR = 1.000 at 5r and 0.996 at 10r, DVR = 0.008/0.024) and, at 10 robots, attains the highest completion rate in every scenario (CR = 0.996–1.000); the consensus and evolutionary baselines match its completion under deadline pressure only by incurring up to $1.9\times$ higher deadline-violation rates (DVR = 0.38–0.49 vs AHE's 0.24). Sub-millisecond decision latency is maintained across all scales (≤ 0.48 ms, ≈ 125–$280\times$ faster than RoSTAM-EA), at the cost of higher makespan and task delay relative to static-commit consensus methods under mixed-stress conditions. In the matched physical campaign at all three scales (180 runs; 10 common seeds per scenario and scale), F58 (P1R) passes the fairness-guarded promotion gate in seven of nine scale–scenario cells with paired-Wilcoxon support: at the primary scale DVR, delay, and geodesic distance fall, the active-robot Jain index rises, and failure recovery shortens by 34 s; the deadline-pressure DVR and distance gains persist at 10 robots, and the two non-passing cells fail only on criteria whose CIs span zero. The four-method Gazebo benchmark is nevertheless produced with the F45 configuration; F58 is reported as a flag-gated, validated refinement that does not alter F45 behaviour. We hope the decomposition into ideal allocation fitness vs. realized Gazebo metrics is useful to other MRTA studies struggling to separate algorithm quality from physical execution noise.

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